



European Economic Interest Group-
European Rail Traffic Management System.
25 Av. de Beaulieu -1160 Brussels - Belgium.
Phone (02) 673-99-33/fax 673-41-50. TVA 455-935.830

Reference EEIG : 98s711
Chapter Ref : 01s7111-

Distribution date :
Document version :

30/09/98
1-

ERTMS/ETCS

RAMS Requirements - Informative part

Chapter 1 - RAM

COMMENT:

In order to avoid confusion between the RAM and Safety references each title of this chapter is prefixed by "R" for RAM.

VERSIONS & MODIFICATIONS

Version Nb	Date of distribution	Comments on the modification	Responsible for the modification
1-	30/09/98	New document derived from 96S1265- Volume 4 due to agreed amendments in the Data Base	M. B.

APPROVAL

This document is approved by :

Title	Name	Signature
RAMS Manager	M. BERIEAU	
Systems Director	F. HEIJNEN	

This document is checked by :

Title	Name	Signature
Quality Director		

This document is authorised for distribution by :

Title	Name	Signature
Managing Director	J. PELLEGRIN	

TABLE OF CONTENTS

R.0. REFERENCES	4
R.1. RAM "STATE OF THE ART" ANALYSIS.....	5
R.1.1 ERRI-A200 RAM STRATEGY	6
R.1.1.1 ERRI-A200 RAM Activity Management.....	6
R.1.1.2 ERRI-A200 RAM Requirements Definition	6
R.1.1.3 Determination of ERRI-A200 probability parameters	8
R.1.1.4 Determination of ERRI-A200 time parameters.....	8
R.1.1.5 Determination of the ERRI-A200 availability target	8
R.1.2 ERRI-A200 OVERALL RAM REQUIREMENTS	9
R.1.2.1 Explicit Overall RAM Requirements from the ERRI-A200 FRS Preliminary.....	9
R.1.2.2 Overall RAM Requirements from the ERRI-A200 RAM Strategy.....	11
R.1.2.3 Necessity of harmonising RAM Requirements.....	11
R.2. Functional RAM Apportionment Methodology	13
R.2.1 FUNCTIONAL APPORTIONMENT FEASIBILITY METHOD	13
R.2.2 QUANTIFIABLE CONTRIBUTION APPORTIONMENT	15
R.2.2.1 Evaluation of HW weighting factors.....	15
R.2.2.2 Apportionment Procedure	17
R.2.3 NOT QUANTIFIABLE CONTRIBUTION APPORTIONMENT	19
R.2.3.1 Functions weakness estimation parameters	19
R.2.3.2 Apportionment Procedure	27
R.2.4 WEIGHTING FACTORS FOR ERTMS HARDWARE.....	28
R.2.4.1 IMMOBILISING and SERVICE failures concerning in ERTMS context.....	28
R.2.4.2 ERTMS equipment typologies.....	28
R.2.4.3 Number of occurrences of ERTMS equipment in 1 hour trip.....	30
R.2.4.4 Equipment typologies weighting factors calculation	32
R.2.4.5 Normalisation of weighting factors with separation of contributions.....	37
R.2.4.6 Normalisation of weighting factors without separation of contributions.....	39
R.3. MIL-HDBK-472 Procedure V	41
R.4. SPECIFICATION OF SERVICE REQUIREMENTS	42
R.4.1 GENERAL REQUIREMENTS	42
R.4.2 SERVICING.....	42
R.4.2.1 General.....	42
R.4.2.2 Construction	42
R.4.2.3 Protection measures	45
R.4.2.4 Maintenance System.....	45
R.4.2.5 Software requirements	48
R.4.2.6 Documentation	48
R.4.3 STANDARDS AND BIBLIOGRAPHY	49
R.5. Specification of environmental requirements	50
R.6. Forecasting on the ERTMS Equipment Targets	51
R.6.1 SYSTEM IDENTIFICATION	52
R.6.2 WORST CASE DISTRIBUTION OF EQUIPMENT IN 1 HOUR TRIP.....	54
R.7. RESULT OF DB AG STUDY - requirements FOR AVAILABILITY OF ERTMS EQUIPMENT	55
ANNEX A: MIL - HDBK - 472 PROCEDURE V - CONTENTS	56
ANNEX B: RESULT OF DB AG STUDY - REQUIREMENTS FOR AVAILABILITY OF ERTMS EQUIPMENT - CONTENTS	101

R.0. REFERENCES

- | | |
|-------------------------|--|
| [R.1] EN 50126 | "Railway Applications - The Specification and Demonstration of Dependability, Reliability, Availability, Maintainability and Safety (RAMS)", CENELEC |
| [R.2] EN 50128 | "Railway Applications - Software for Railway Control and Protection Systems",
CENELEC |
| [R.3] ENV 50129 | "Railway Applications - Safety Related Electronic Systems",
CENELEC |
| [R.4] UIC/ERRI | "ETCS - FRS Functional Requirements Specification",
Version 4.01 |
| [R.5] UIC/ERRI | "ETCS RAM Strategy",
Final Draft, 28 July 1995 |
| [R.6] MIL-HDBK-338-1A | "Electronic Reliability Engineering Handbook", Vol. 1,
USA DoD, 12 October 1988 |
| [R.7] DB AG STUDY | "Result of DB AG Study - Requirements for Availability on ERTMS Equipment" |
| [R.8] ERTMS USERS GROUP | Environmental Conditions, 97s0665- |

R.1. RAM "STATE OF THE ART" ANALYSIS

This preliminary RAM analysis has the purpose of examining the *ERRI-A200 ETCS* documentation, in order to recognise the fundamental concepts which the overall RAM requirements have to be based on.

In this activity the *ETCS ERRI-A200* documentation is reviewed in order to extract and to isolate all the RAM-related information in the context of the required system performance.

The final goal is to provide a baseline for harmonising, as far as possible, the *ERTMS Control/Command* system RAM requirements, formalised as recommended by the international standards [R.1], [R.2] and [R.3], with the other system requirements and with the RAM requirements defined by *ERRI-A200*.

The following outputs constitute the preliminary RAM activities results:

1. **Explicit RAM constraints.** Explicit RAM targets defined at the system level must be located and isolated to be formalised in the Overall RAM Requirements Specification.
2. **Implicit RAM constraints.** Implicit RAM constraints may be hidden in the functional specifications of the system. They must be disclosed and, if necessary, translated into appropriate RAM requirements in this preliminary phase.

The *ERRI-A200* terminology will be adopted in the following paragraphs, so the term *ETCS*, belonging to the *ERRI-A200* glossary, will be utilised for referring the *ERTMS Control/Command* system.

R.1.1.1 ERRI-A200 RAM STRATEGY

This paragraph has the aim of discussing the basic concepts presented in the *ERRI-A200* document *ETCS RAM Strategy* in order to:

- identify the current level of understanding as far as *ETCS* RAM clauses are concerned;
- suggest and provide modifications to be adopted with respect to the above document.

R.1.1.1.1 ERRI-A200 RAM Activity Management

It is possible to subscribe to the first part of the document as far as the RAM process, organisation and planning are concerned, in that the basic criteria are correctly appointed. It is anyway necessary to develop that concepts in order to include, in a formal way, the RAM Programme Requirements in the Overall RAM Requirements Specification.

In particular, the RAM strategy states the following:

- the *ETCS* RAM process shall be carried on in compliance with [2.1];
- the *ETCS* RAM Programme must be planned by means of the RAM Programme Plan (RPP) which shall be clearly and univocally defined including descriptions of:
 - the RAM Organisation
 - the relationships among the *ETCS* RAM Programme and Quality Plan;
 - the RAM activities to be carried on along the whole *ETCS* lifecycle;
 - the responsibilities in developing RAM activities;
 - the RAM activities time planning.

In order to accomplish the above needs, the *ETCS* RAM Programme will be specified in the RAM Requirements Specification for all the phases of the *ETCS* lifecycle phases: Design, Implementation, Installation, Test and Operation.

The response from the field shall be a Overall RAM Programme Plan integrating the individual RAM Programme Plans defined, according to the RAM Programme Specification, by the organisations responsible for each *ETCS* lifecycle phase.

R.1.1.1.2 ERRI-A200 RAM Requirements Definition

Some modifications are necessary in the definition of the system availability targets. In particular, the basic principles adopted for treating software reliability need to be revised. Furthermore, according to the failure conditions earlier defined, the re-

quirements have to be differentiated in reason of their impact on the train circulation.

For example, as far as this second aspect is concerned, the alternatives of repairing a train on site or of requiring a depot level intervention should be formalised according to separated availability requirements.

The first aspect, related to software reliability, is instead a rather delicate problem. As the international standards, like [R.2], and the current experience recommend, no quantitative requirements have to be stated for software.

Nevertheless, software failures are reasonably expected to cause unavailabilities of programmable electronic systems.

The problem is how to link operational requirements, which must be quantitatively expressed, with merely qualitative requirements for software items.

In this section a methodology for defining operational requirements and for assigning upper limits to the contribution due to the system items, in reason of their nature, is presented according to the basic indications provided by the *ERRI-A200* document.

The first step is to determine an operational availability target for the system. To this aim, the procedure adopted in the *ERRI-A200* document appears acceptable, but the proposed numerical values have to be critically revised.

The proposed procedure follows these steps:

R.1.1.3 Determination of ERRI-A200 probability parameters

1. to assess the current probability of having a trip delay for generic causes;
2. to assess the probability of having delay because of technical causes;
3. to assess the probability of having delay caused by signalling failures;
4. to state the acceptable probability of having delay due to *ETCS* failures: P_{ds} ;

R.1.1.4 Determination of ERRI-A200 time parameters

5. to state the acceptable average value for the delay, concerning only delayed trips: T_{dy} ;
6. to assume an average duration of the *ETCS* failure time for each delayed trip (it is assumed equal to $0.9 \cdot T_{dy}$ in the *ERRI-A200* document);
7. to assess the average trip normal duration for delayed trips only: T_{dnd} ;
8. to assess the average trip normal duration for **not** delayed trips only: T_{dn} ;

R.1.1.5 Determination of the ERRI-A200 availability target

9. to define the *ETCS* operational availability target:

$$A_o = \frac{T_{op}}{T_{op} + T_{fault}} = \frac{1 - P_{ds} T_{dn} + P_{ds} T_{dnd}}{1 - P_{ds} T_{dn} + P_{ds} T_{dnd} + P_{ds} \cdot 0.9 \cdot T_{dy}}$$

Once the operational availability A_o is defined, the unavailability time over one year of operation T_u is fixed. Starting from this value it is possible to determine the diverse failure contributions to the system unavailabilities.

The contributions to the system unavailabilities may be identified in:

- Software contribution;
- Hardware contribution;
- Transmission contribution.

Only for HW and for Transmission are defined quantitative target which have to be demonstrated by analysis and test.

For SW are defined only qualitative requirements according to the level of criticality of the functions to be implemented.

The procedure adopted is described in the following paragraphs.

Starting from the acceptable average operational unavailability time over one year of operation, the maximum contribution due to HW and Transmission is stated, for example as the 60% of the operational unavailability.

The resulting value is apportioned in reason of the 90% for HW and of the 10% for Transmission. This fact results in the definition of numerical availability targets to be demonstrated by means of an appropriate RAM Programme.

As can be noted no upper level for the SW-related unavailabilities is defined and so no one specific quantitative requirement is stated for software.

This approach is necessary in order to guarantee that the operational availability target is reached: in phase of demonstration, the failures due to HW and Transmission will be charged to the relevant specific quantitative requirements while the failures due to SW defects will be charged directly to the operational availability target.

The software qualitative targets will be expressed in terms of integrity levels, as the approach proposed in [R.2] suggests, corresponding to different levels of applications of the SW Quality Plan and assigned to the system functions in reason of their criticality.

R.1.2 ERRI-A200 OVERALL RAM REQUIREMENTS

The purpose of this section is to recognise, in the current documentation, each reference to RAM Requirements for *ETCS*.

The following documents have been analysed:

- *ERRI-A200 ETCS* FRS Preliminary [R.4];
- *ERRI-A200 ETCS* RAM Strategy [R.5].

The objective is to provide basic considerations for discussing, modifying, harmonising and, if necessary, eliminating the above references.

R.1.2.1 Explicit Overall RAM Requirements from the ERRI-A200 FRS Preliminary

The *A200 FRS Preliminary* defines constraints concerning RAMS requirements which should become inputs for the Overall RAM (and S) Requirements Specification.

The *FRS Preliminary* states some figures for the RAM requirements to be used as starting point for the complete RAM Specification.

The figures established by A200, to be discussed and justified, are the following:

R.1.2.1.1 Availability

Overall availability target: 99,98%

R.1.2.1.2 Maintainability

– Onboard equipment

Max allowed time to detect a module failure:	5 s
Max allowed time to replace a faulty module:	5 s
Max time to restart the system:	15 s
Max additional time to replace a traction unit:	3 hours

– Trackside equipment

Max allowed time to detect an equipment failure:	5 s
Max allowed time to inform maintenance personnel:	5 min
Max allowed time to reach the equipment failed:	2 hours
Max allowed time to replace an equipment:	
<i>Balise</i>	5 min
<i>Loop</i> [*]	10 min
<i>Antenna</i>	10 min

^{*} Loop electronic equipment

R.1.2.2 Overall RAM Requirements from the ERRI-A200 RAM Strategy

The *A200 ETCS RAM Strategy* defines, on the basis the procedure earlier mentioned, the following numeric values to be revised and discussed:

R.1.2.2.1 Probability parameters

Probability of having a trip delay for generic causes: = 15%

Probability of having delay because of technical causes:

$$60\% \cdot 15\% = 9\%$$

Probability of having delay caused by signalling failures:

$$40\% \cdot 9\% = 3,6\%$$

Probability of having delay due to *ETCS* failures:

$$P_{ds} = 40\% \cdot 3,6\% = 1,54\%$$

R.1.2.2.2 Time parameters

Average value for the delay (only delayed trips):

$$T_{dy} = 20 \text{ min}$$

Average duration of the *ETCS* failure time for a delayed trip:

$$= 0,9 T_{dy}$$

Average trip normal duration:

$$T_{dnd} = T_{dn} = 90 \text{ min}$$

R.1.2.2.3 Availability target

The availability target is determined by applying the formula:

$$A_o = \frac{T_{op}}{T_{op} + T_{fault}} = \frac{1 - P_{ds} T_{dn} + P_{ds} T_{dnd}}{1 - P_{ds} T_{dn} + P_{ds} T_{dnd} + P_{ds} \cdot 0,9 \cdot T_{dy}} = \frac{90}{90 + 0,0154 \cdot 0,9 \cdot 20} = 0,99693$$

which represents the *operational availability* target for *ETCS*.

R.1.2.3 Necessity of harmonising RAM Requirements

The targets defined as input for the Overall RAM Specification must be harmonised in order to be effectively developed in the RAM specification activities. In particular, the following needs have been recognised:

- the availability target derived from the *ERRI-A200* FRS Preliminary has to be justified and the type of availability of concerning has to be clearly indicated (operational, inherent, achieved ...);

- the maintainability targets derived from the *ERRI-A200* FRS Preliminary are the only cited, so they require to be discussed and justified;
- the assumptions made for deriving the operational availability target in the *ERRI-A200* RAM Strategy have to be discussed and, if necessary, revised.

Once the above activities are carried out, the harmonisation of the RAM inputs for the ERTMS/ETCS Overall RAM Requirements Specification can be achieved.

R.2. Functional RAM Apportionment Methodology

This part aims at defining the basic methodology to be utilised for apportioning the *ERTMS Control/Command* Overall RAM Requirements to the functions defined in [R.4].

The methodology is composed of two steps:

1. apportionment of the quantifiable contribution due to hardware and transmissions: this step is based on the estimation of the complexity of the hardware liable to implement the *ERTMS Control/Command* functions;
2. apportionment of the not quantifiable contribution due to software: this step is based on the estimation of the weakness of the *ERTMS Control/Command* functions themselves intended as the liability of causing system failures due to software defects;

The main problem to be faced for performing the first step is that the *ERTMS Control/Command* hardware architecture is, currently, not defined yet and then the hardware-based functional apportionment must only refer to the operating principles defined in the § 2.2.1.4. of the Overall RAM Requirements Specification, in addition, obviously, to the *ERTMS Control/Command* FRS defining the functions themselves.

In the context of this approach it is important to identify the typology of the system components for evaluating the feasibility of the RAM Requirements to be apportioned to the *ERTMS Control/Command* functions.

As far as the second step is concerned, it is important to identify the correct parameters both significant and exhaustive for the functions complexity estimation, but, in the same time, easily applicable starting from the FRS.

The purpose of this second step is to identify an IL for each of the *ERTMS Control/Command* function taking into account the following two aspects: the function criticality and the function weakness. In this way it is possible to link the IL with the concept of software reliability.

R.2.1 FUNCTIONAL APPORTIONMENT FEASIBILITY METHOD

The diagram in Figure 1 summarises the philosophy adopted by the proposed apportionment method, named *Functional Apportionment Feasibility Method*, as applied to the *ERTMS Control/Command* system.

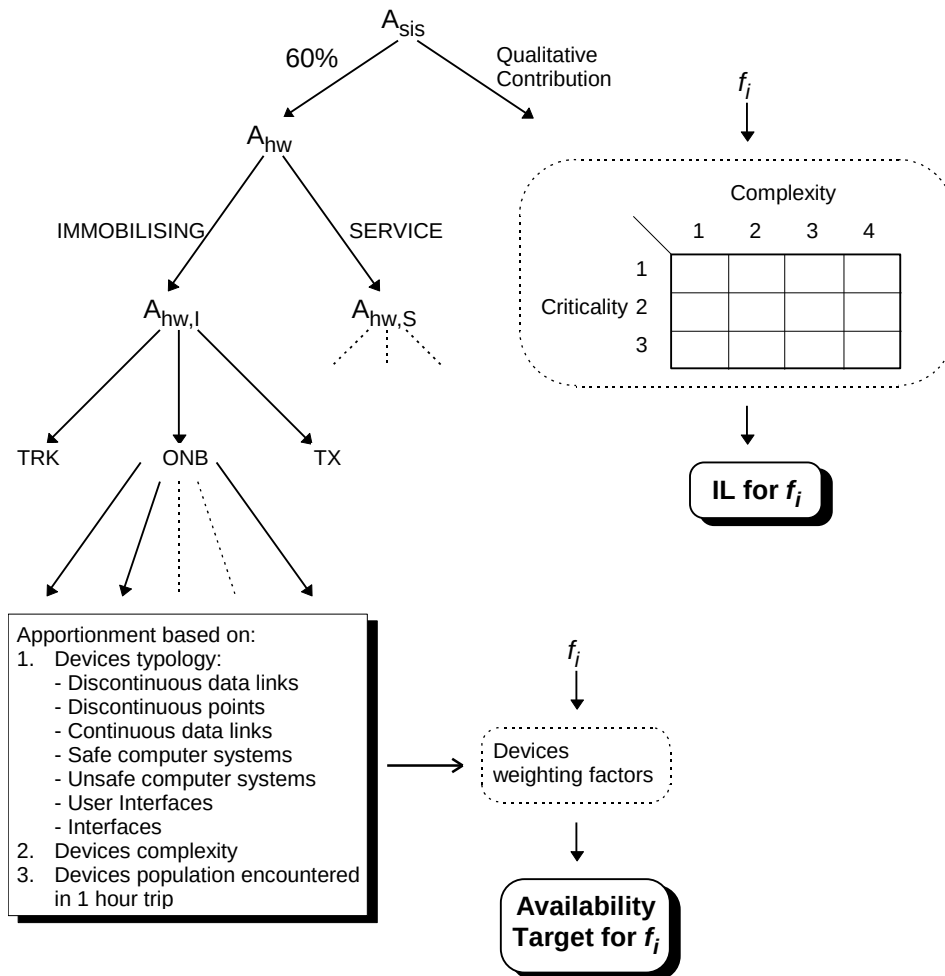


Figure 1 - The Functional Apportionment Feasibility Method

R.2.2 QUANTIFIABLE CONTRIBUTION APPORTIONMENT

R.2.2.1 Evaluation of HW weighting factors

The basis of this approach is the FEASIBILITY OF OBJECTIVES apportionment methodology proposed by [R.6].

This technique was initially developed for allocating reliability without repair in electrical-mechanical systems. In this context, it has been adjusted for apportioning availability, in terms of downtime, to the *ERTMS Control/Command* functions by determining appropriate allocation factors computed for the hardware devices which implement them.

The allocation factors represent, in the original technique as described by [R.6], a function of numerical ratings of system intricacy, state of the art, performance time and environmental conditions. For apportioning availability instead of reliability, further parameters, relevant to the devices maintenance-related aspects and to the presence of redundancy, are necessary: the ease of maintenance and the type of redundancy.

Each rating is on a scale from 1 to 10, except for the redundancy rating scaled from 1 to 100, and is estimated by the engineer on the basis of his experience:

1. *System Intricacy*. Intricacy is evaluated by considering the probable number of parts or components making up the system and also is judged by the assembled intricacy of these parts or components. The least intricate system is rated as 1 and a highly intricate system is rated at 10.
2. *State of the Art*. The state of present engineering progress in all fields is considered. The least developed design or method is assigned a value of 10 and the most highly developed is assigned a value of 1.
3. *Performance Time*. The element that operates for the entire mission time is rated 10 and the element that operates the least time during the mission is rated at 1.
4. *Environment*. Environmental conditions are also rated from 10 through 1. Elements expected to experience harsh and very severe environments during their operation are rated as 10, and those expected to encounter the least severe environments are rated as 1.

5. *Ease of maintenance*. The elements most easily repairable are rated as 1 and the least easily repairable are rated as 10. The ease of maintenance must take into account the time necessary for reaching the maintenance location.
6. *Redundancy*. Elements expected to be implemented without redundancy are assigned a value of 100 while elements fully redounded with repair of the faulty units are rated as 1.

The six ratings are multiplied to give the subsystem rating which will range between 1 and 10^7 .

The above ratings are assigned, in the *ERTMS Control/Command* context, to the devices, categorised by typology, encountered by a train during a 1 hour trip.

The following typologies of devices are recognised in the above context:

- discontinuous data links;
- discontinuous reset points;
- continuous data links;
- safe computer systems;
- unsafe computer systems;
- user interfaces;
- system interfaces.

The devices weighting factors will be determined once known the number of items for each typology encountered during a 1 hour train trip and once known the different contributions assigned to trackside, onboard and transmission equipment.

Finally, it will be possible to determine the apportionment factors by normalising the above defined ratings so that their sum is 1 with respect to the assumed operating scenario.

R.2.2.2 Apportionment Procedure

The following procedure is adopted for allocating the appropriate availability targets to the *ERTMS Control/Command* functions:

1. Isolate the contribution to the system availability, A_{hw} , due to hardware failures: this is quantified in the 60% of A_{sys} .
2. Subdivide A_{hw} among IMMOBILISING and SERVICE failures according to appropriate weighting factors, as an example:
$$(1-A_{hw,I}) = 10\% (1-A_{hw})$$
$$(1-A_{hw,S}) = 90\% (1-A_{hw})$$
3. Identify the typologies of devices which concur to the above unavailabilities.
4. Identify the *ERTMS Control/Command* operating conditions in terms of the number of devices, and the relevant typology, encountered by a train during a 1 hour trip (on the basis of the § 2.2.1.4.) differentiated by Application Level, if applicable.
5. Apply the *Functional Apportionment Feasibility* method on the above devices tacking into account the different contributions due to Trackside, Onboard and Transmission equipment on system unavailabilities. The result is a set of normalised weighting factors for the different device typologies:

$\{w_1, w_2, \dots, w_N\}$ where N is the number of different device typologies

6. For each *ERTMS Control/Command* function, f_i , identifies the set of the devices, in terms of their typology, necessary for its success:

$\{D_1, D_2, \dots, D_{Ni}\}$ where N_i is the number of devices vital for f_i

7. Identify the functions criticality: IMMOBILISING, SERVICE, MINOR.
8. Compute the downtimes corresponding to the availability targets:

$$\begin{array}{ll} A_{hw,I} & \rightarrow \quad DT_{hw,I} \\ A_{hw,S} & \rightarrow \quad DT_{hw,S} \end{array}$$

9. Determine the downtime requirement, and subsequently the availability target, for each *ERTMS Control/Command* function:

$$f_{I,i} \Rightarrow DT_{f_{I,i}} = \left(\sum_{j=1}^{N_i} r_{i,j} \cdot w_j \right) \cdot DT_{hw,I} \Rightarrow A_{f_{I,i}}$$
$$f_{S,k} \Rightarrow DT_{f_{S,k}} = \left(\sum_{j=1}^{N_k} r_{k,j} \cdot w_j \right) \cdot DT_{hw,S} \Rightarrow A_{f_{S,k}}$$

where $r_{i,j}$ represents the percentage of utilisation of the device j by the function i .

10. Determine an availability target for MINOR criticality functions and apply the formula defined at point 9.:

as an example

$$A_{hw,M} = 0.995 \quad \rightarrow \quad DT_{hw,M}$$
$$f_{M,l} \Rightarrow DT_{f_{M,l}} = \left(\sum_{j=1}^{N_l} r_{l,j} \cdot w_j \right) \cdot DT_{hw,M} \Rightarrow A_{f_{M,l}}$$

For the application of this method it is important that a person with a deep understanding of the *ERTMS Control/Command* system provide all the necessary information requiring appropriate technical skills. The success of the apportionment is conditioned to the effectiveness of the interface between the system technical specialist and the RAM Engineer.

R.2.3 NOT QUANTIFIABLE CONTRIBUTION APPORTIONMENT

The purpose of this section is to identify a procedure for allocating an appropriate IL to the software implementing the *ERTMS Control/Command* functions on the basis of RAM considerations.

The approach is entirely qualitative and based on parameters derivable from the appropriate software documentation.

Two factors affect the process of determining the functions IL:

1. the functions criticality in terms of their effect on the system operation;
2. the functions weakness in terms of structural and environmental weakness parameters.

The criticality of the *ERTMS Control/Command* functions has been already discussed in § R.2.2.

As far as the functions weakness is concerned, the Structural Weakness parameters relates to the probability of injecting defects in the software modules during the software development process, while the Context Weakness parameters provide an estimation of the probability that such defects become effective during the function execution and then cause a system failure.

R.2.3.1 Functions weakness estimation parameters

The following parameters are defined in order to estimate the weakness of the software implementing *ERTMS Control/Command* functions.

The set of the above parameters derives from the experience in other similar software project development, and each of them can assume qualitative values (*low, medium, high* for instance) whose assignment is performed on the basis of system and software engineering judgement based on experience or on comparative considerations.

Parameters evaluation is qualitative, because their value is expressed in terms of adimensional measures, such as: High, Low, Medium, deriving from metrics evaluation. Metrics are evaluated in terms of adimensional and qualitative measures too, depending on physical and environmental conditions of the software code, which will implement *ERTMS Control/Command* function.

For each metric, the minimum and the maximum values shall be defined by the extreme cases, resulting from analysis of the software which will implement the *ERTMS Control/Command* function under review, on the basis of engineering judgement, deriving, when possible, from experience in similar applications.

R.2.3.1.1 Structural Weakness Parameters

1. Size (SZ)

[1 Low, $10 \div 10^2$ Medium, $10^3 \div 10^4$ High]

The size related factors of the code implementing functions influence directly the number of expected defect. The estimated code length depends on the total number of operators and on the total number of operands. The estimated code volume depends on the number of unique operators and on the number of unique operands.

Metrics:

- Estimated Code Length (SZ1)
[1 Low, 10 Medium, 10^2 High]
- Estimated Code Volume (SZ2)
[1 Low, 10 Medium, 10^2 High]

2. Control Flow Complexity (FC)

[1 Low, 10^2 Medium, 10^4 High]

Control Flow Complexity represents a measure of the logical function complexity, influencing strongly the introduction of defects. This parameter is not necessarily related to the program size. Estimated cyclomatic number may be evaluated on the basis of the function control graph.

Metric:

- Estimated Cyclomatic Number
[1 Low, 10^2 Medium, 10^4 High]

3. Data Complexity (DC)

[1 Low, 10² Medium, 10⁴ High]

The Data Complexity of a module implementing a function is evaluated as the sum of the complexities of the data structures used in the module. The evaluation of the Data Complexity is based on the assumption that functions which manage large and complex data set are more liable to present defects. Therefore the relevant metric may be the Data Structure Complexity.

Metric:

- Data Structure Complexity (DS)
[1 Low, 10² Medium, 10⁴ High]

4. Fault Tolerance Features (FT)

[1 Low, 10 Medium, 10² High]

Fault Tolerance Features, built in the software architecture, reduce software defects influence on the function operational effectiveness.

Techniques:

- Recovery Blocks
[1 Not Adopted, 10 Adopted]
- N-Version Programming
[1 Not Adopted, 10 Adopted]

5. Defects Reduction (DR)

[1 Low, 10÷10² Medium, 10³÷10⁴ High]

Defects Reduction want to give a measure of the development effort dedicated to the fault avoidance and fault removal.

Metrics:

- Process Development Maturity
[1 Low, 10 Medium, 100 High]
- Test Coverage
[1 Low, 10 Medium, 100 High]

R.2.3.1.2 Context Weakness Parameters**1. Time Features (TF)**

[1÷10 Low, 10²÷ 10⁴ Medium, 10⁵÷10⁸ High]

Time Features are strictly related to the function time stress. This environmental weakness parameter influences the probability that a defect in the software implementing a function results in the function failure due to time constraints.

Metrics:

- Function Operative Mode
[1 Off-Line, 10 On-Line, 100 Real-Time]
- Dynamic Performance
[1 Low, 10 Medium, 100 High]
- Task Number
[1 Low, 10 Medium, 100 High]
- Multitasking Features
[1 No, 100 Yes]

2. Human Factor Features (HF)

[1÷10 Low, 10²÷10³ Medium, 10⁴÷10⁶ High]

Human Factor Features aim at emphasise the importance of the human component on the function success.

Metrics:

- Human Interaction Mode
[1 No Interactive, 10 Low Interactive, 10² High Interactive]
- Required Operator Skill (1 in absence of human interaction)
[1 Low, 10 Medium, 10² High]
- Degree of Novelty (1 in absence of human interaction)
[1 Low, 10 Medium, 10² High]

3. Occurrences Level (OC)

[1 Low, 10^4 Medium, 10^6 High]

The number of occurrences influences naturally the function operative success. This parameter gives a measure of the number and of the duration of the function calls in the context of the whole system operational profile.

Metrics:

- Percentage of Function Execution in Total System Execution
[1 Low, 10^4 Medium, 10^6 High]

Example: Metric Application - Determining the Structural factor Size (SZ)

Input

Estimated total number of operators = $N1 = 100$

Estimated total number of operands = $N2 = 50$

Estimated number of unique operators = $n1 = 100$

Estimated number of unique operands = $n2 = 120$

Judgement Criteria

These criteria relate on the physical measures performed on the software and the qualitative ones. They shall be derived on the basis of engineering judgement, deriving, when possible, from experience in similar applications. For example, for the specified programming language, we may have the following judgement criteria.

For $N1+N2 \leq 200$ SZ1 = Low

For $200 < N1+N2 \leq 350$ SZ1 = Medium

For $N1+N2 > 350$ SZ1 = High

For $n1+n2 \leq 150$ SZ2 = Low

For $150 < n1+n2 \leq 200$ SZ2 = Medium

For $n1+n2 > 200$ SZ2 = High

Procedure

With the input values and the criteria we are able to calculate SZ1 and SZ2:

$N1+N2 = 150 \Rightarrow$ SZ1 is Low = 1

$n1+n2 = 220 \Rightarrow$ SZ2 is High = 100

Output

Finally, we can obtain the value of SZ:

$SZ = \text{Size} = SZ1 \cdot SZ2 = 1 \cdot 100 = 100 \Rightarrow$ SZ is Medium

R.2.3.1.3 Weakness Graph

The attribution of the appropriate weakness class to the *ERTMS Control/Command* functions is made by means of a qualitative method named *Weakness Graph*.

Four weakness classes, from 1 to 4, are defined. The class 1 is assigned to the most robust function while the class 4 is assigned to the most weak one.

The additional class 0 is assigned to the functions whose IL depends *exclusively* on their criticality.

The path to be followed in the Weakness Graph for the generic function f_i is determined according to the three factors defined in the following table (Table I):

Table I		
Factor	Value	Source
Expected Number of Defects (Nd)	Nd3 High ($10^8 \div 10^{12}$)	Structural Weakness Parameters SZ · FC · DC
	Nd2 Medium ($10^3 \div 10^7$)	
	Nd1 Low ($1 \div 10^2$)	
Defect-caused Failures Frequency (Fd)	Fd3 High ($10^{12} \div 10^{20}$)	Context Weakness Parameters TF · HF · OC
	Fd2 Medium ($10^3 \div 10^{11}$)	
	Fd1 Low ($1 \div 10^2$)	
Defects and Failures Reduction (Rd)	Rd1 Absent (1)	Fault Tolerance Features Developers Experience Test Effort FT · DR
	Rd2 Low ($10 \div 10^3$)	
	Rd3 High ($10^4 \div 10^6$)	

In Figure 2 the Weakness Graph for assigning the appropriate weakness class to the function f_i is presented.

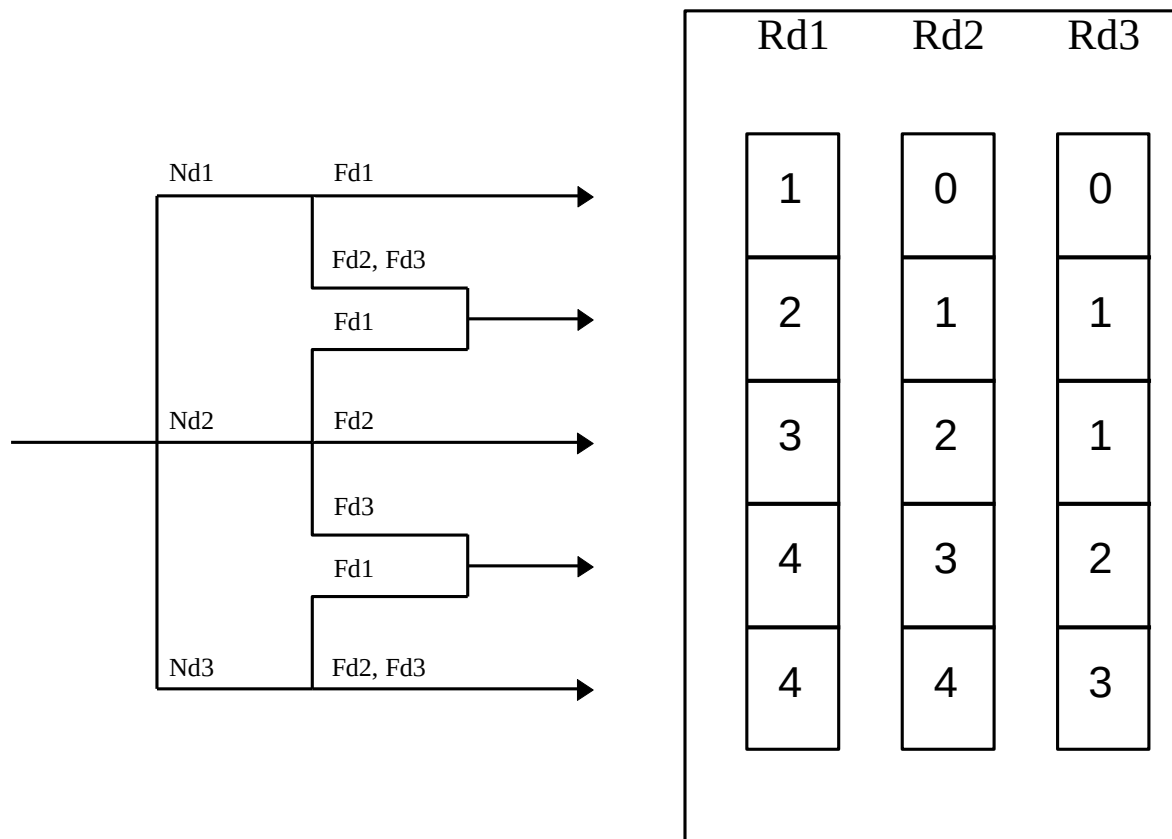


Figure 2 - The Weakness Graph

Example: Weakness Graph Application

Input

We have for the function N the following input data:

Nd = Nd2

Fd = Fd3

Rd = Rd1

Procedure

Step1: we enter in the Weakness Graph (see Figure 3) through the Nd2 branch.

Step 2: we leave the Weakness Graph through the Fd2 branch. We obtain the following Weakness Class values:

- 4 for Rd = Rd1;
- 3 for Rd = Rd2;
- 2 for Rd = Rd3.

Step 3: we obtain the Weakness Class, on the basis of the defined Defects and Failure Reduction Rd1, as 4.

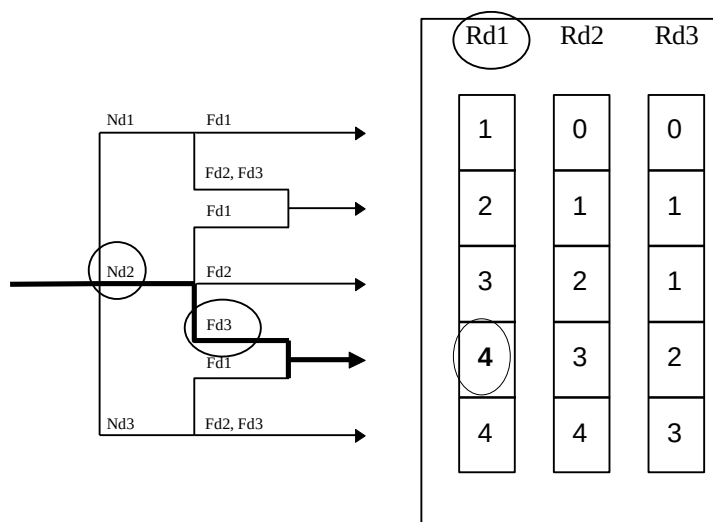


Figure 3 - Weakness Graph application: an example

Output

Function Weakness Class = 4.

After it has been defined function f_i weakness class and its criticality, using the matrix presented in Figure 3, the function integrity level IL can be derived.

This matrix correlates the failure effects and the probability of failure appearance, and is called *Failure Risk Matrix*.

Weakness Class → Criticality ↓	WC0	WC1	WC2	WC3	WC4
0 - NOT RELEVANT	IL0	IL0	IL0	IL0	IL0
1 - MINOR	IL0	IL0	IL1	IL1	IL1
2 - SERVICE	IL0	IL1	IL1	IL2	IL2
3 - IMMOBILISING	IL1	IL1	IL2	IL2	IL3

Example: Failure Risk Matrix Application

We have a function with a criticality of 2 and a Weakness Class of WC1, using the Failure Risk Matrix, we obtain IL2 as its Integrity Level.

Weakness Class → Criticality ↓	WC0	WC1	WC2	WC3	WC4
0 - NOT RELEVANT	IL0	IL0	IL0	IL0	IL0
1 - MINOR	IL0	IL0	IL1	IL1	IL1
2 - SERVICE	IL0	IL1	IL1	IL2	IL2
3 - IMMOBILISING	IL1	IL1	IL2	IL2	IL3

R.2.3.2 Apportionment Procedure

The following procedure is adopted for allocating the appropriate IL to the *ERTMS Control/Command* functions, based on RAM implications:

1. Assign the appropriate Weakness Class to the function f_i on the basis of the weakness parameters defined in § R.2.3.1.
2. Recognise the function f_i criticality on the basis of the consequences of its failure (see point 7. of § R.2.2.2.).
3. Assign the appropriate IL to the function f_i on the basis of the Failure Risk Matrix defined in § R.2.3.1.

R.2.4 Weighting Factors for ERTMS Hardware

R.2.4.1 IMMOBILISING and SERVICE failures concerning in ERTMS context

The definition of IMMOBILISING and SERVICE failures provided in the § 2.1.1.5. of the *ERTMS Control/Command* RAM Requirements Specification needs to be tailored for the ERTMS context.

In fact, any *ERTMS Control/Command* failure cause a vehicle to be unable to move, but some ERTMS failures may be as severe as that condition.

In the context of *ERTMS Control/Command* the failure condition definitions to be referred can be found in the Glossary.

The following failure conditions are taken into account:

- *Service failure*
- *Immobilising failure*
- *Minor failure*

R.2.4.2 ERTMS equipment typologies

In this section will be discussed the typologies of equipment recognised in the context of the *ERTMS Control/Command* system for implementing the *ERTMS Control/Command* functions as defined in the A200-FRS document.

Only the mission critical devices, as described in the EEIG ERTMS Users Group document "Degraded Modes", are considered in this report.

R.2.4.2.1 OnBoard Equipment

The following equipment typologies are necessary for implementing the *ERTMS Control/Command* onboard functions:

1. *OnBoard Safe Module* - implements the *safety related* on board functions;
2. *OnBoard Non-Safe Module* - implements the *not safety related* on board functions;
3. *User Interface* - implements the driver interfacing functions;
4. *System Interfaces* - implements the interfacing functions with external OnBoard subsystems or with other ERTMS OnBoard subsystems:
 - Odometer Interface;
 - Train Interface;

- Continuous Communication Interface;
 - Intermittent Communication Interfaces;
5. *Radio Equipment* - realise the interface to the GSM Equipment for providing the physical connection for continuous train-to-track and track-to-train communications;
 6. *Spot RX/TX Equipment* - realise the physical connection for intermittent track-to-train communications;

R.2.4.2.2 Trackside Centralised Equipment

The following equipment typologies are necessary for implementing the *ERTMS Control/Command* centralised trackside functions:

1. *Trackside Safe Module* - implements the *safety related* trackside functions;
2. *Trackside Non-Safe Module* - implements the *not safety related* trackside functions;
3. *System Interfaces* - implements the interfacing functions with external Trackside systems (Regulation and Interlocking) or with other ERTMS Trackside subsystems (other centralised equipment, track devices, continuous communication system):
 - Regulation and Interlocking Interface;
 - Continuous Communication Interface;
 - Trackside Distributed Equipment Interface;
 - Trackside Centralised Equipment Interface;
4. *Radio Equipment* - realise the interface to the GSM Equipment for providing the physical connection for continuous train-to-track and track-to-train communications and optional connections with other ERTMS Trackside equipment.

The above equipment will be hereinafter simply called *Trackside Equipment*.

R.2.4.2.3 Trackside Distributed Equipment

The following equipment typologies are necessary for implementing the *ERTMS Control/Command* distributed trackside functions:

1. *Intermittent Communication Points* - provide spot information to trains, comprise:
 - Control Unit;
 - Active Transducers;
2. *Reset Points* - provide spot information for the train odometer reset, comprise:
 - Passive Transducers.

The above equipment will be hereinafter called *Lineside Equipment*.

R.2.4.3 Number of occurrences of ERTMS equipment in 1 hour trip

An estimation of the number of equipment, for each typology, encountered by a train within a 1 hour trip is carried out in this section.

R.2.4.3.1 OnBoard Equipment

The following equipment are liable to form the OnBoard equipment of an *ERTMS* equipped train (for Level 3, worst case):

- 1 Safe Module (OSM);
- 1 Non-Safe Module (OUSM);
- 1 User Interface (OUI);
- 4 Simple System Interfaces (OSI - Continuous Communication, Intermittent Communications and Odometer);
- 1 Complex System Interface (OCSI - Train Interface);
- 2 Spot RX/TX Apparatus (OIRT - Loop and Balise);
- 1 Radio Equipment (ORE).

R.2.4.3.2 Trackside Centralised Equipment

The following equipment are liable to form the Trackside equipment:

- 1 Safe Module (TSM);
- 1 Non-Safe Module (TUSM);
- 2 Simple System Interfaces (TSI - Interlocking & Regulation and other Trackside equipment);
- 2 Complex System Interface (TCSI -Trains and Lineside equipment);
- 1 Radio Equipment (TRE)

R.2.4.3.3 Trackside Distributed Equipment

The following equipment are liable to form the Lineside equipment:

Intermittent Communication Point

1. Information Point
 - 2 Active Transducers (LAT);
 - 1 Control Unit (LCU);
2. Entry/exit Point
 - 8 Active Transducers (LAT);
 - 1 Control Unit (LCU);

Reset Point

- 1 Passive Transducer (LPT).

R.2.4.3.4 Worst case distribution of equipment in 1 hour trip

As mentioned in § 2.2.1.4.1. the following equipment are encountered by a train in a 1 hour trip:

1	Train	1	OSM
		1	OUSM
		1	OUI
		4	OSI
		1	OCSI
		2	OIRT
		1	ORE
10	Trackside Centralised Equipment	10	TSM
		10	TUSM
		20	TSI
		20	TCSI
		10	TRE
440	Intermittent Com Points	1120	LAT (2 x 400 + 8 x 40)
		440	LCU (400 + 40)
500	Reset points	1000	LPT

R.2.4.4 Equipment typologies weighting factors calculation

On the basis of engineering judgement, based on experience on similar systems, the following weighting factors are assigned to the *ERTMS Control/Command* Equipment typologies.

Onboard Safe Module - OSM	
<i>Factor</i>	<i>Value</i>
System Intricacy	8
State of the Art	1
Performance Time	10
Environment	8
Ease of Maintenance	6
Redundancy	100
Total	$3.84 \cdot 10^5$

Onboard Non-Safe Module - OUSM	
<i>Factor</i>	<i>Value</i>
System Intricacy	6
State of the Art	1
Performance Time	10
Environment	8
Ease of Maintenance	6
Redundancy	100
Total	$2.88 \cdot 10^5$

Onboard User Interface - OUI	
<i>Factor</i>	<i>Value</i>
System Intricacy	6
State of the Art	1
Performance Time	10
Environment	8
Ease of Maintenance	6
Redundancy	100
Total	$2.88 \cdot 10^5$

Onboard Simple System Interface - OSI	
<i>Factor</i>	<i>Value</i>
System Intricacy	3
State of the Art	3
Performance Time	10
Environment	8
Ease of Maintenance	4
Redundancy	100
Total	$2.88 \cdot 10^5$

Onboard Complex System Interface - OCSI	
<i>Factor</i>	<i>Value</i>
System Intricacy	5
State of the Art	3
Performance Time	10
Environment	8
Ease of Maintenance	5
Redundancy	100
Total	$6 \cdot 10^5$

Onboard Spot RX/TX Apparatus - OIRT	
<i>Factor</i>	<i>Value</i>
System Intricacy	3
State of the Art	3
Performance Time	2
Environment	8
Ease of Maintenance	3
Redundancy	100
Total	$4.32 \cdot 10^4$

Onboard Radio Equipment- ORE	
<i>Factor</i>	<i>Value</i>
System Intricacy	5
State of the Art	5
Performance Time	10
Environment	8
Ease of Maintenance	5
Redundancy	100
Total	$1 \cdot 10^6$

Trackside Safe Module - TSM	
<i>Factor</i>	<i>Value</i>
System Intricacy	10
State of the Art	1
Performance Time	10
Environment	1
Ease of Maintenance	1
Redundancy	10*
Total	$1 \cdot 10^3$

Trackside Non-Safe Module - TUSM	
<i>Factor</i>	<i>Value</i>
System Intricacy	6
State of the Art	1
Performance Time	10
Environment	1
Ease of Maintenance	1
Redundancy	10*
Total	$6 \cdot 10^2$

Trackside Simple System Interface - TSI	
<i>Factor</i>	<i>Value</i>
System Intricacy	3
State of the Art	2
Performance Time	10
Environment	1
Ease of Maintenance	1
Redundancy	10*
Total	$6 \cdot 10^2$

Trackside Complex System Interface - TCSI	
<i>Factor</i>	<i>Value</i>
System Intricacy	5
State of the Art	2
Performance Time	10
Environment	1
Ease of Maintenance	1
Redundancy	10*
Total	$1 \cdot 10^3$

Trackside Radio Equipment - TRE	
<i>Factor</i>	<i>Value</i>
System Intricacy	5
State of the Art	5
Performance Time	10
Environment	1
Ease of Maintenance	1
Redundancy	10*
Total	$2.5 \cdot 10^3$

Lineside Passive spot Transducer - LPT	
<i>Factor</i>	<i>Value</i>
System Intricacy	1
State of the Art	2
Performance Time	2
Environment	8
Ease of Maintenance	4
Redundancy	100
Total	$1.28 \cdot 10^4$

Lineside Active spot Transducer - LAT	
<i>Factor</i>	<i>Value</i>
System Intricacy	2
State of the Art	2
Performance Time	2
Environment	8
Ease of Maintenance	4
Redundancy	100
Total	$2.56 \cdot 10^4$

Lineside Control Unit - LCU	
<i>Factor</i>	<i>Value</i>
System Intricacy	3
State of the Art	1
Performance Time	10
Environment	8
Ease of Maintenance	4
Redundancy	100
Total	$9.6 \cdot 10^4$

* It has been assumed that each Trackside Centralised Equipment can be operatively replaced by the adjacent one in case of failure. This involve the presence of hot redundancy.

R.2.4.5 Normalisation of weighting factors with separation of contributions

In this approach, the weighting factors carried out in § R.2.4.4. require to be normalised in such a way that their sum is 1.

Additional assumptions, which can be interpreted as constraints for the RAM Apportionment, are made before to proceed in the normalisation process:

1. The contribution of OnBoard and Trackside Equipment to the *ERTMS Control/Command* HW failures is the same (50% and 50%);
2. Trackside Centralised Equipment and Trackside Distributed Equipment, as a whole, contribute in the same way to Trackside HW failures (50% and 50% then 25% and 25% on the total contribution to *ERTMS* HW failures).

This involve that the OnBoard equipment weighting factors sum will be equal to 0.5, the Trackside weighting factors sum will be equal to 0.25 and the Lineside weighting factors sum will be 0.25 as well.

R.2.4.5.1 OnBoard contribution

The weighting factors are normalised in such a way that their sum, tacking into account the number of items involved for each typology, is 0.5.

$$1 \times \text{OSM} + 1 \times \text{OUSM} + 1 \times \text{OUI} + 4 \times \text{OSI} + 1 \times \text{OCSI} + 1 \times \text{ORE} + 1 \times \text{OIRT}$$

↓

$$1 \times 3.84 \cdot 10^5 + 1 \times 2.88 \cdot 10^5 + 1 \times 2.88 \cdot 10^5 + 4 \times 2.88 \cdot 10^5 + 1 \times 6 \cdot 10^5 + 1 \times 10^6 + 2 \times 4.32 \cdot 10^4 = \mathbf{3.8 \cdot 10^6}$$

$$\text{Normalisation Factor } K = 0.5 / 3.8 \cdot 10^6 = \mathbf{1.32 \cdot 10^{-7}}$$

$$\text{OSM}_n = K \cdot \text{OSM} = 5.07 \cdot 10^{-2}$$

$$\text{OUSM}_n = K \cdot \text{OUSM} = 3.8 \cdot 10^{-2}$$

$$\text{OUI}_n = K \cdot \text{OUI} = 3.8 \cdot 10^{-2}$$

$$\text{OSI}_n = K \cdot \text{OSI} = 3.8 \cdot 10^{-2}$$

$$\text{OCSI}_n = K \cdot \text{OCSI} = 7.92 \cdot 10^{-2}$$

$$\text{ORE}_n = K \cdot \text{ORE} = 1.32 \cdot 10^{-1}$$

$$\text{OSM}_n = K \cdot \text{OSM} = 5.7 \cdot 10^{-3}$$

R.2.4.5.2 Trackside contribution

The weighting factors are normalised in such a way that their sum, tacking into account the number of items involved for each typology, is 0.25.

$$10 \times \text{TSM} + 10 \times \text{TUSM} + 20 \times \text{TSI} + 20 \times \text{TCSI} + 20 \times \text{TRE}$$

$$\Downarrow$$

$$10 \times 10^3 + 10 \times 6 \cdot 10^2 + 20 \times 6 \cdot 10^2 + 20 \times 10^3 + 10 \times 2.5 \cdot 10^3 = \mathbf{7.3 \cdot 10^4}$$

$$\text{Normalisation Factor } K = 0.25 / 7.3 \cdot 10^4 = \mathbf{3.4 \cdot 10^{-6}}$$

$$\text{TSM}_n = K \cdot \text{TSM} = 3.4 \cdot 10^{-3}$$

$$\text{TUSM}_n = K \cdot \text{TUSM} = 2.08 \cdot 10^{-3}$$

$$\text{TSI}_n = K \cdot \text{TSI} = 2.08 \cdot 10^{-3}$$

$$\text{TCSI}_n = K \cdot \text{TCSI} = 3.4 \cdot 10^{-3}$$

$$\text{TRE}_n = K \cdot \text{TRE} = 8.5 \cdot 10^{-3}$$

R.2.4.5.3 Lineside contribution

The weighting factors are normalised in such a way that their sum, tacking into account the number of items involved for each typology, is 0.25.

$$1120 \times \text{LAT} + 1000 \times \text{LPT} + 440 \times \text{LCU}$$

$$\Downarrow$$

$$1120 \times 2.56 \cdot 10^4 + 1000 \times 1.28 \cdot 10^4 + 440 \times 9.6 \cdot 10^4 = \mathbf{8.37 \cdot 10^7}$$

$$\text{Normalisation Factor } K = 0.25 / 8.37 \cdot 10^7 = \mathbf{3 \cdot 10^{-9}}$$

$$\text{LPT}_n = K \cdot \text{LPT} = 3.84 \cdot 10^{-5}$$

$$\text{LAT}_n = K \cdot \text{LAT} = 7.68 \cdot 10^{-5}$$

$$\text{LCU}_n = K \cdot \text{LCU} = 2.88 \cdot 10^{-4}$$

R.2.4.6 Normalisation of weighting factors without separation of contributions

Also in this second approach, the weighting factors carried out in § R.2.4.4. require to be normalised in such a way that their sum is 1.

In this case the weighting factors are normalised in such a way that the whole sum is one. In a second step, the operational HW availability will be separated in two contributions, which define apportionment constraints, due to IMMOBILISING and SERVICE failures respectively.

When allocating the availability target to an *ERTMS Control/Command* function, the weighting factors of the function belonging equipment will multiply the appropriate availability contribution in reason of the criticality of the function (IMMOBILISING or SERVICE).

$$1 \times \text{OSM} + 1 \times \text{OUSM} + 1 \times \text{OUI} + 4 \times \text{OSI} + 1 \times \text{OCSI} + 1 \times \text{ORE} + 2 \times \text{OIRT} + 10 \times \text{TSM} + 10 \times \text{TUSM} + 20 \times \text{TSI} + 20 \times \text{TCSI} + 10 \times \text{TRE} + 1120 \times \text{LAT} + 1000 \times \text{LPT} + 440 \times \text{LCU}$$

$$\Downarrow$$

$$\begin{aligned} & 1 \times 3.84 \cdot 10^5 + 1 \times 2.88 \cdot 10^5 + 1 \times 2.88 \cdot 10^5 + 4 \times 2.88 \cdot 10^5 + 1 \times 6 \cdot 10^5 + \\ & + 1 \times 10^6 + 2 \times 4.32 \cdot 10^4 + 10 \times 10^3 + 10 \times 6 \cdot 10^2 + 20 \times 6 \cdot 10^2 + 20 \times 10^3 + \\ & + 10 \times 2.5 \cdot 10^3 + 1120 \times 2.56 \cdot 10^4 + 1000 \times 1.28 \cdot 10^4 + 440 \times 9.6 \cdot 10^4 \cong \\ & \cong 8.76 \cdot 10^7 \end{aligned}$$

$$\text{Normalisation Factor } K = 1 / 8.76 \cdot 10^7 \cong 1.14 \cdot 10^{-8}$$

$$\text{OSM}_n = K \cdot \text{OSM} = 4.38 \cdot 10^{-3}$$

$$\text{OUSM}_n = K \cdot \text{OUSM} = 3.29 \cdot 10^{-3}$$

$$\text{OUI}_n = K \cdot \text{OUI} = 3.29 \cdot 10^{-3}$$

$$\text{OSI}_n = K \cdot \text{OSI} = 3.29 \cdot 10^{-3}$$

$$\text{OCSI}_n = K \cdot \text{OCSI} = 6.85 \cdot 10^{-3}$$

$$\text{ORE}_n = K \cdot \text{ORE} = 1.14 \cdot 10^{-2}$$

$$\text{OIRT}_n = K \cdot \text{OSM} = 4.93 \cdot 10^{-4}$$

$$\text{TSM}_n = K \cdot \text{TSM} = 1.14 \cdot 10^{-5}$$

$$\text{TUSM}_n = K \cdot \text{TUSM} = 6.85 \cdot 10^{-5}$$

$$\text{TSI}_n = K \cdot \text{TSI} = 6.85 \cdot 10^{-5}$$

$$\text{TCSI}_n = K \cdot \text{TCSI} = 1.14 \cdot 10^{-5}$$

$$\text{TRE}_n = K \cdot \text{TRE} = 2.85 \cdot 10^{-5}$$

$$\text{LPT}_n = K \cdot \text{LPT} = 1.46 \cdot 10^{-4}$$

$$\text{LAT}_n = K \cdot \text{LAT} = 2.92 \cdot 10^{-4}$$

$$\text{LCU}_n = K \cdot \text{LCU} = 1.10 \cdot 10^{-3}$$

This second approach presents two advantages:

- it is possible to state a constraint on the maximum incidence of the failures categorised as IMMOBILISING and SERVICE respectively;
- the real complexity of the ERTMS HW structure is taken into account.

R.3. MIL-HDBK-472 Procedure V

This part recalls the principles defined by MIL-HDBK-472 Procedure V for Maintainability Predictions, in the same way they are presented in that military handbook.

The contents of this Part are completely identical to those present in MIL-HDBK-472, so it can be used as informative reference in substitution of the original document without limitations.

This part is copied in Annex of this document.

R.4. SPECIFICATION OF SERVICE REQUIREMENTS

This document is the result of a DB AG Study, constituting the basis for the Maintenance requirements developed in the chapter 02s1266- of 96s1266-.

R.4.1 GENERAL REQUIREMENTS

This document describes the general requirements for maintenance and servicing of ETCS trackside and on board equipment. If no other stipulations are made below, in principle, the requirements relating to the construction and maintenance of electronic and technical safety installations stipulated in European Standards as e.g. EN 50128, EN 50155 and IEC 321 have to be applied.

R.4.2 SERVICING

R.4.2.1 General

During all stages of design, particular emphasis must be placed on reduced wearing, a high level of stress capability, cost-effective servicing and technical treatment over extended intervals in short periods. Particular reference is made to the requirement for all system, hardware and software designs to comply with EN ISO 9001 with clearly specified functional and interface specifications. As far as useful and necessary in the design concept has to be included protective measures against terrorism and safety related servicing failures. When constructing ETCS installations it must be ensured that it is possible to undertake any unscheduled repairs (excluding accident-related repairs) within a maximum standstill period of 4 hours by on board and trackside equipment and 2 hours by ETCS equipment within central towers.

Note:

By unscheduled repairs the replacement of closed component groups in the engine area or under the vehicle has to be enclosed within the standstill period of utmost 4 hours.

R.4.2.2 Construction

R.4.2.2.1 Modular construction

According to functionality and considering economical aspects the ETCS installations has to be of a modular construction. The separate testing of modules should be possible.

The fixing system for these modules and their electrical connections must be such that the modules can be exchanged quickly and simply if required. Each module must be equipped with suitable lifting points. Doors, covers and flaps (where possible with equivalent dimensions) which cover the components or devices to be examined, must be fitted with simple and safe quick release locks. Access to parts carrying voltages in excess of 1000 V must be charged by using special locking mechanisms.

Parts, to which access or repair is particularly difficult must be designed to guarantee a working life of at least 20 years without the need for major servicing or repair resulting from corrosion or ageing. This applies particularly to the cabling.

Note:

The requirement for a problem-free and rapid replacement procedure extends also to the equipment/ devices contained in the modules.

R.4.2.2.2 Mounting and arrangement of equipment

The installations must be arranged, secured and distributed so that they are easily accessible, controllable and replaceable as required, without damage to and unacceptable interference with other components or wiring. In the case of locally fixed installations directly in the area of the platform, the clearance profile stipulated by the operator must be complied with.

Note:

In the range of DB-AG additionally to the GC clearance profile, stated in UIC leaflet 506 the areas for the purposes of operating and servicing of railway lines has to be kept free.

Mounting of ETCS-equipment on the frame of passenger vehicles has to consider the movements of modern passenger vehicles with greater suspension ways during normal operation in vertical and (also inclined) horizontal direction.

The general design and also the arrangement of the installations must take into account the fact that repair work of components is not generally carried out locally. Therefore, all components, particularly for work carried out in repair shops, must be easily accessible and replaceable, as far as possible without having to remove adjacent parts.

If, in justified exceptional cases, pre-adjustment to balance the operation is considered to be necessary (i.e. no internal calibration), the adjustment elements must be accessible during operation of both the entire and adjacent installations. These adjustment elements must retain their setting during normal operation and be protected against unintentional modification.

R.4.2.2.3 Housings

The installations must be housed to ensure that they can be operated under the specified operating conditions.

The housings must be designed to

- guarantee protection of the installations under operating conditions and the respective possible environmental/local conditions and external influences (e.g. mechanical or electrical influences, contamination, temperature and humidity etc.)
- facilitate problem-free removal or repair of the installations contained within it
- where possible, hinder or prevent the installations being affected by sabotage

Note:

Frames, component group carriers and plug installations must conform to Standard IEC 297.

R.4.2.2.4 Identification of components and installations

Component group carriers and fitted printed circuit boards must be adequately labelled so that they can be correctly identified, including serial number and alteration. All labels must be clear, concise and permanent. Labels affixed to replaceable installations must also provide designation, name of manufacturer or trade name and the serial number. Wherever possible, the component lettering must be visible on the installed component.

Note:

Component group carriers and fitted printed circuit boards must have the means to show alterations to the installation, form or function. Where possible, the identification plate must be arranged on the front plate of plug installations. In the case of printed circuit boards, the image must have sufficient details to facilitate correct identification and also revision.

R.4.2.2.5 Interfaces

All interfaces must be designed so that the installation fulfils its requirement under the respective operating and environmental conditions.

Note:

Fulfilling the requirements of the interfaces considers also the limitation of the consequential damage caused by external errors, particularly in relation to:

- *electromagnetic compatibility*
- *potential differentials*
- *safety of persons*

For the interfaces between modules, the transfer procedure and also the information to be transferred must be clearly defined including logical and also physical value ranges. Wherever possible and feasible, the various modules of ETCS installations must be galvanically separated from each other.

R.4.2.2.6 Components

All components used must be produced to a Quality System in accordance with the requirements of EN 29001 or EN 29002 or an equivalent Quality System and - quoting the references - correspond to international standards and specifications such as IEC, which specify the functional and physical parameters of the components.

The quality of the components used has to guarantee perfect functioning of the installation under any possible conditions of use which may arise. The choice of components and component groups has to ensure that after delivering of the installation there is every probability of further consignments over an extended period. Before the end of the useful life of the installation, replacement types which are equivalent in terms of function and operation for components which become unavailable must be stipulated.

Note:

Where possible, components has to be used which can be supplied functionally and operationally equivalent by several manufacturers. In case that components with only one supplier cannot be avoided, this has to be justified and the operator has to be informed at the time of tendering.

Special components such as customer-specific hybrid switches and application-specific integrated switching circuits must be described with adequate precision in a specification, in order to allow subsequent development or to serve as a source of information for fully compatible components from alternative suppliers.

R.4.2.2.7 Signalling and operating installations

Where appropriate indicators to illustrate the condition of the installation (e.g. input values, output values, control functions, power supplies) must be used to support a diagnosis.

The results of the diagnosis must be illustrated separately for the various user groups (e.g. vehicle driver, maintenance staff), preferably with pictures. The arrangement of the pictures has to be organised hierarchically.

To select the information appropriate input options must be allocated to the displays, which facilitate a controlled, rapid and unproblematic manual selection of the required information, i.e. indication of the results of the diagnosis, operating instructions or in case of a failure, a limitation of non-diagnosable defects.

Displays in driver cabins must be designed and installed so that all the information (text, graphics) can easily be read off under any light or lighting conditions (night/sunlight) in accordance to technical standards. To ensure easily and safely read-off of the display in the drivers-cabin statistical or dynamic displays with an image repetition frequency of at least 100 images/sec. should be used. For displays in drivers-cabins in vehicles, hierarchically structured images should preferably be selected by means of soft-key keyboard.

Note:

For a safe and perfect read-off of the display a contrast strength of at least 1:160 must be maintained. Image resolution must be at least 640 x 480 Pixel. It must be possible to read-off the display at an angle of 40° to the horizontal and 20°/25°¹ to the vertical. The display must be resistant to incidental light; to adjust to the various light conditions, the dimmer range must be adjustable from 3 Cd/m² to 300 Cd/m².

R.4.2.2.8 Internal flexible wiring (electrical and optical)

It must be guaranteed that for securing, fixing methods (bolts, clamps etc.) appropriate for the anticipated power load both during normal operation and also in the case of a short circuit are used (see also Section R.4.2.3.4 Fault Protection).

It must be possible to follow all wiring by using a connection diagram or list.

Wires which could be subject to movement, must be equipped with appropriate terminals or supports near to the point of connection and at suitable points along the length. The wires must be arranged such that their operating behaviour is not affected by high temperatures. It is necessary to arrange the wiring according to the mechanical load.

Note:

The wires must not be bent to radii smaller than the smallest permissible value to be specified by the manufacturer. Although no minimum radius is stipulated for electrical cable and lines, the internal radius of the curve must not be smaller than the external diameter of the wire, including its insulation sheath. Glands or bushes must be used if the wiring is inserted through any type of material suspected to cause damages through wearing.

Wherever practically possible, a sufficient length of wire should be used so that it is possible to place a new connection at each end of the wire. Shielded cable and wires must have an insulating external coating.

The conductor cross section must be chosen for the respective circuit so that in the event of a power surge, the wires do not sustain damage before the protection device is triggered.

¹ from top/bottom

R.4.2.3 Protection measures

R.4.2.3.1 General

For the production, commissioning and operation of ETCS installations, the respective valid national and international standards must be complied with.

R.4.2.3.2 Fire protection

The requirements for the fire protection of ETCS installations are based on the requirements of the appropriate Fire Protection Category which applies for the vehicle and/or the section installation in accordance with DIN 5510 and DIN EN 2. With respect to the requirements for fire protection measures for installations and rail vehicles, Standard prEN45545 must be complied with.

In the event of fire, in order to avoid the occurrence of highly aggressive and toxic compounds which subsequently necessitate extensive disposal measures, only materials which do not have these side effects may be used.

Note:

The connection of ETCS-installations to outside on-board-equipments has to be done by halogenfree cables.

R.4.2.3.3 Protection in the event of indirect contact

The equipments of ETCS installations has to be designed so that no dangerous contact voltages can occur or, where, in the event of a failure, the occurrence of a dangerous contact voltage cannot be excluded, the necessary measures are taken to ensure physical protection.

R.4.2.3.4 Fault protection

The installations must be protected against external faults (e.g. short circuit or interruption conditions). All protection devices used must be arranged so as to minimise the risk of fire within the installation.

Note:

Outgoing cables must be dimensioned at least for the threshold voltage of the protection device. Controlled voltage supply sources for electronic installations must contain a voltage limiter to reduce the need for safety fuses.

Where protection devices with triggers are contained in the output voltage circuit, the voltage available under short circuit conditions must be sufficiently high to actuate the same. Moreover, devices which are re-set manually, must be easily accessible.

R.4.2.4 Maintenance System

R.4.2.4.1 General

In order to ensure fast repair, an adequate quantity of replacement parts must be available. On completion of any repair work, a full functional test must be performed through the System. Re-start is only possible after proof of proper functioning.

R.4.2.4.2 Maintenance documentation

The maintenance system for the ETCS installations must be matched to the proposed operational use and include all details necessary for maintenance scheduling and implementation and also for staff training, production of appliances and the procurement of testing and measuring equipment. An appropriate training programme which can be run preferably on IBM compatible PCs must be produced for training and re-training of maintenance staff.

R.4.2.4.3 Diagnostic and testing equipment

R.4.2.4.3.1 General

For all functional groups, a fast and reliable fault detection and limitation system must be provided by means of diagnostic and testing equipment, which both during the connecting phase and also during continuous operation, tests the installation automatically (self-diagnosis) and gives a clear indication of the operating condition of the installation.

The diagnostic and test equipment should limit the fault to the smallest replaceable installation or - where this is not possible - to the damaged individual function. By signalling selected messages and referring to the measures to be taken in particular cases, in the event of a fault, the diagnostic and testing equipment should serve to support the operating and maintenance staff. The stored fault list must be sufficiently comprehensive to record all maintenance related faults. It must be possible to expand the list.

All fitted diagnostic facilities which, in addition to signals, can also trigger test conditions, must be locked in appropriate manner to prevent an interruption of the correct operation of the installation outside of the test mode. For subsequent fault analysis, the fault conditions and the accompanying information necessary for the analysis must be stored in non-volatile memories.

In addition to the self-diagnosis, including all sub-systems and signalling of the fault during operation, approved test runs must be available for easier maintenance through operation, including those covering safety functions, operator elements and auxiliary devices, indication of the test results and - where necessary - recording of test data.

R.4.2.4.3.2 Diagnostic concept

The aim of the diagnosis is to increase the availability of the system and to minimise both maintenance and Life-Cycle costs.

The diagnosis must therefore:

- facilitate quality and expenditure control of the maintenance systems
- in the event of a fault, support the drive vehicle staff by means of signals and references to the technical operation measures to be taken in particular cases
- enable the workshop staff to execute a fast and economic fault correction in the event of a fault by indicating through special references the smallest installation replaceable from a technical and economic point of view,
- save faults / diagnostic results together with date, time and other circumstances and/or operating data in non-volatile memories
- carry a current list of all possible diagnostic messages
- where technically possible, suppress diagnostic messages based on consequential faults
- output support measures and operating instructions separately for each user group, e.g. vehicle driver
- initiate automatic test runs to check operational readiness
- modifications of functions defining elements (e.g. Q , R_{iso} , f , ...) has to be identified and to indicate a service request even before the reaching the limits (failure). The limits shall be adjustable by trained maintenance staff

The diagnostic results should have the following quality features (after completion of one year of operation) in accordance with IEC 56 CO 160:

recognisable proportion of faults: 98%
reliability of diagnosis: 95%

Diagnosis of the ETCS installations must be carried out predominantly automatically and wherever possible, regardless of the operating condition. Tests which can affect the operating behaviour of the vehicle must be initiated by the vehicle driver (e.g. forced braking test).

The functional state of the ETCS System and the degree of urgency for repairs in the event of a fault must be signalled to the operators (e.g. next stop or end of journey). In the event of a fault within the ETCS installation, the fault must be signalled to a central point so that the action required to rectify the fault, e.g. provision of the necessary spare parts, can be taken.

R.4.2.4.3.3 Structure of the diagnostic installations

To carry out a diagnosis, the self-diagnosis method should predominantly be used, whereby the functional group controlling the process also carries out its diagnosis. The diagnosis should therefore, wherever possible use the existing functional data and the sensory mechanism already used for the function. Additional

sensory mechanisms may only be installed if they can be justified either from a safety point of view or are necessary to achieve the diagnostic objective. The use of additional structural elements for fitted diagnostic installations must not significantly affect the reliability of the installation and must be taken into account for the reliability calculations. In the event of a fault, the diagnostic installation must be able to precisely identify the primary cause and suppress the signalling of consequential faults on the same installation.

Fundamentally, all ETCS System components must be integrated into the diagnosis and be equipped with a service interface for communication with external facilities (e.g. PC) which also provide operating functions. The fault limiting programs and test programs required for this form part of the specification for the respective ETCS installation.

The installations must be constructed so as to facilitate maintenance and diagnostic measures and also - where necessary linking in external test equipment - without removing, replacing, damaging or inappropriately affecting the components or wiring of the installation being examined. The installations must be designed so that a failed, replaceable installation can be identified either by means of a suitable portable test device or a fitted diagnostic installation, both with attendant test instructions. Moreover, installed printed circuit boards must have test points, e.g. test plugs, test connection surfaces etc., which support the diagnosis and repair sequence.

Regardless of the type and degree of the operational restrictions, the diagnosis must indicate whether the repair work has to be carried out immediately or as part of the normal maintenance schedule and/or whether repair work of on board equipment has to be done in the workshop.

For the purpose of subsequent fault analysis, the nature of the fault, including date, time and the respective operating data must be stored in a non-volatile archive memory which can be read out and evaluated via a service interface. It must only be possible for authorised persons to have access to and delete diagnostic data relating to a ETCS installation. Appropriate tools form part of the diagnosis. The data memory must be dimensioned and organised in such a way that no data can be lost, especially relating to vehicles between workshop visits.

Note:

As standard interfaces for connecting external testing and diagnostic installations, should be used the RS 232 Interface with a 9 pin Submin D plug in accordance with IEC 807-2. For the specific diagnosis through maintenance, the necessary software must be supplied by the supplier of the electronic control system. A standardised user surface must be guaranteed. Signals based on the so-called "loose connection" must be recognised in the appliance and dealt with accordingly. The so called „workshop mode“ should enable the workshop staff to initiate test routines for the most important components, starting from the display and either to obtain the results as displayed on screen and/or read out via a PC. It should only be possible to use this mode for test purposes which should be restricted to normal operation. During maintenance work, the diagnostic data should be marked for diagnostic evaluation.

R.4.2.5 Software requirements

R.4.2.5.1 General

The Software must comply with the requirements stipulated in standards EN50128 and EN50155.

R.4.2.6 Documentation

R.4.2.6.1 General

For the execution of maintenance and also modification and revision work, complete documentation for each component of the ETCS System must be handed to the operator. These documents must conform to the requirements of EN 50155 - in the case of Software, also the requirements of EN 50128 - and must be stated in a comprehensive list with details of their current processing and revision status.

The documents must have an appropriate drawing number and a title describing the particular illustrated part and the type of drawing. All documents and structural component lists must have an edition and revision code and index for modification.

Notes:

The following list should be used as a starting point for the minimum documentation required:

- a) *Name, type and functional purpose of the installation*
- b) *Structure of the entire installation and mode of operation*
- c) *Instructions for installation and adjustment of values*
- d) *Description of switching mode, where applicable, including the curve form of the voltage and the power, voltage rise time etc.*
- e) *Description of the functional interface*
- f) *status of modifications/changes*
- g) *Planning and production documents so far as necessary for maintenance.
(e.g. block switching images, circuit diagrams, wiring and connection diagrams, drawings and photographs, adjustment and measuring sheets...)*
- h) *Diagnostic methods and required test installations*
- i) *Precautionary methods for storage*
- j) *Approved functional switching diagram*
- k) *Layout plans, drawings and photographs for mechanical arrangement*
- l) *component lists*
- m) *Lists of structural components with limited lifespan*
- n) *component specifications and delivery details (i.e. manufacturer)*
- o) *Testing and measuring points*
- p) *Details of dangerous substances which may be present within the installation and which have been approved by the user*
- q) *Details of possible risks of implosion or explosion existing within the installation or which may occur during use or handling*

R.4.2.6.2 Maintenance Manual

All documents relating to maintenance must be organised in one manual and should describe the following points:

- individual functions to be carried out (where necessary, supported by diagrams) and details on frequency/intervals
- any special tools, appliances, testing and measuring equipment required to carry out the work
- the time required to carry out the work in terms of both individual functions and as a whole for the respective maintenance stage
- the operating materials and energy types to be used (compressed air, alternating and direct current etc.)
- pre-conditions of the system

R.4.2.6.3 Software and Documentation

The Software must be documented clearly and simply so that a test, maintenance and subsequent adjustment/changes to the Software are possible. Software documentation has to be elaborated in accordance with the requirements stipulated in EN50128 and EN50155.

R.4.3 STANDARDS AND BIBLIOGRAPHY

See References in the “ERTMS Users Group - ERTMS/ETCS Environmental Conditions” ref 97s066

See also the Normative References in “ERTMS Users Group -RAMS Requirements specifications” ref 96s126

Without laying claim to completeness, the following lists contains the titles of further relevant standards and references:

UIC 506	Regulations for the Application of Expanded Limitation Lines GA, GB and GC
CENELEC PrEN45545	Railway applications - Fire protection on Railway vehicle.

R.5. Specification of environmental requirements

This part has been intentionally deleted because it is currently replaced by the following mandatory document :

ERTMS/ETCS Environmental Conditions Ref : 97s066 Version 5-

R.6. Forecasting on the ERTMS Equipment Targets

This part aims at performing a forecasting about the MTBF values which could be required for the *ERTMS Control/Command* subsystems and/or equipment.

At the current detail level of the *ERTMS Control/Command* RAM Requirements Specification, limited to the functional apportionment of the Operational Availability overall target, what can be indicated in this context is a conservative forecasting based on the Functional analysis and on the functional FMEA carried out in Chapter 2 - paragraph 2.3. in 96s126.

For this reason, the MTBF estimated values, reported in Table R.6.I, must be read in a very critical way taking into account the following hypotheses:

- the values indicated are referred only to the SERVICE failure condition as defined in chapter 0 - Glossary of 96s126;
- the system is considered in his Worst Case conditions travelling at a speed of 500 km/h;
- all the items encountered by the train over 1 hour of trip are considered in series configuration for the mission success.

Then, the results presented in Table R.6.I represent what it is possible to be required, as *ERTMS Control/Command* equipment MTBF, in the worst possible case.

R.6.1 SYSTEM IDENTIFICATION

Aim of this paragraph is to briefly recall the architecture of the *ERTMS Control/Command* as the reference architecture for the MTBF forecasting.

As mentioned in Chapter 2 of 96s126- §2.2.1.1.3., the target system is defined, in the RAM Requirements Specification context, as follows:

- 1 *ERTMS Control/Command* equipped train plus all the *ERTMS Control/Command* trackside and lineside equipment encountered during 1 hour of trip in the worst case (at the maximum allowed speed in european railways of 500 km/h considering the most complex possible configuration);

The first target is therefore to identify the extent of the interested items.

The *ERTMS Control/Command* is composed of the following two main categories of equipment:

- *Trainborne equipment:* has the aim of supervising vehicle operation according to the information received from infrastructure installations, from other non ERTMS onboard equipment, from the driver and from the trackside signalling system. The Trainborne equipment includes the following elements:
 - *ERTMS Control/Command* Onboard Safe Module composed by the European Vital Computer (EVC);
 - *ERTMS Control/Command* Onboard UnSafe Module composed by the Management Computer (MC) and by the Recording Unit (RU);
 - Onboard User Interface composed by the Man Machine Interface (MMI);
 - Interface to Odometer Sensors composed by the Time and Odometer Unit (TOU);
 - Interfaces to the Intermittent RX/TX equipment (OSI);
 - Interface to the Onboard Radio Equipment (OSI);
 - Onboard Intermittent RX/TX equipment composed by EuroBalise and Euroloop RX/TX equipment (OIRT);
 - Onboard Radio Equipment composed by the Radio Interface Module, excluding GSM Equipment (ORE);

- Train Interface Unit (OCSI);

– *Trackside equipment:*

has the aim of exchanging information with the vehicle for safely supervising train circulation. The information exchanged between track and trains can be either continuous or discontinuous according to the ERTMS Level of Application and to the nature of the information itself. The trackside equipment may be either distributed along the line, renamed in this case as *Lineside* elements, or centralised. Trackside equipment includes the following elements:

Lineside elements

- Lineside Active Transducers or Switchable Balise;
- Lineside Passive Transducers or Not Switchable Balise;
- Lineside Control Unit or Logical Electronic Units (LEUs);
- Loop identifiable like a Lineside Active Transducer.

Trackside elements (centralised)

- Radio Block Centre (RBC);

R.6.2 WORST CASE DISTRIBUTION OF EQUIPMENT IN 1 HOUR TRIP

As mentioned in the part R.2 of this document, the following equipment are encountered by a train in a 1 hour trip:

1	Train	1 OSM 1 OUSM 1 OUI 4 OSI 1 OCSI 2 OIRT 1 ORE
10	Trackside Centralised Equipment	10 TSM 10 TUSM 20 TSI 20 TCSI 10 TRE
440	Intermittent Com Points	1120 LAT (2 x 400 + 8 x 40) 440 LCU (400 + 40)
500	Reset points	1000 LPT

Table R.6.I

		Weight	Total DT [h]	Allocated DT [h]	Availability	MTTR [h]	MTBF [h]
Train	Onboard Functions	4,34E-02	1,150	0,04987	0,999994307	1,737	3,05E+05
	OSM	4,38E-03	1,150	0,00504	0,999999424	1,737	3,02E+06
	OUSM	3,29E-03	1,150	0,00378	0,999999568	1,737	4,02E+06
	OIRT	4,93E-04	1,150	0,00057	0,999999935	1,737	2,68E+07
	ORE	1,14E-02	1,150	0,01313	0,999998501	1,737	1,16E+06
	OUI	3,29E-03	1,150	0,00378	0,999999568	1,737	4,02E+06
	OCSI	6,85E-03	1,150	0,00788	0,999999101	1,737	1,93E+06
	OSI	3,29E-03	1,150	0,00378	0,999999568	1,737	4,02E+06
Discontinuous	Lineside Functions	9,56E-01	1,150	1,09917	0,999874524	1,737	1,38E+04
	LPT	1,46E-04	1,150	0,00017	0,999999981	1,737	9,05E+07
	LAT	2,92E-04	1,150	0,00034	0,999999962	1,737	4,53E+07
	LCU	1,10E-03	1,150	0,00126	0,999999856	1,737	1,21E+07
Radio	Redounded RBC Functions	8,33E-04	1,150	0,00010	0,999999989	0,435	3,97E+07
	RBC	-	-	1,29569	0,999852090	0,869	5,87E+03

R.7. RESULT OF DB AG STUDY - requirements FOR AVAILABILITY OF ERTMS EQUIPMENT

This part presents the study provided by DB AG on possible ERTMS equipment.

This part is copied in Annex of this document.

This result and the result of R6 have contributed to establish the table of (un)availability targets on constituents in the mandatory part of the RAMS Requirements Specification (ERTMS Users Group : 96s1266- Chapter 2)

.

ANNEX A: MIL - HDBK - 472 PROCEDURE V - CONTENTS

1. GENERAL

This procedure is primarily used to predict maintainability parameters of avionics and ground and shipboard electronics at the organizational, intermediate and depot levels of maintenance. It can also be applied to any application environment and type of equipment including mechanical equipment.

1.1 Philosophy and assumptions. The maintainability prediction procedures presented herein permit the maintainability of electronic equipment/systems to be analyzed, including direct accountability of diagnostics, isolation, and test capabilities; replaceable item (RI)² construction; packaging; and component failure rates. In addition, the following assumptions and stipulations apply to any predictions made using the procedures:

- a. Failure rates experienced are all in the same proportion to those predicted
- b. Only one failure at a time is considered
- c. Maintenance is performed in accordance with established maintenance procedures
- d. Maintenance is performed by maintainers possessing the appropriate skills and training
- e. Only active maintenance time is addressed; administrative and logistic delays, and clean-up are excluded

Two separate methods are presented. Method A is an prediction method for use in the early stages of the development of an equipment or system. Method B is a detailed prediction method that uses actual detailed design data to predict maintainability parameters.

The application of the procedures presented here permits the user to monitor the overall system maintainability throughout the design and development of that system. The user can identify whether or not the specified maintainability design requirements will be met before the system is complete. Thus, if it appears the maintainability requirements will not be met, the designers can be informed and the necessary changes can be made before they become prohibitively expensive.

1.2 Point of application. Both prediction methods (Method A is the early prediction and Method B is the detailed prediction) of this procedure can be applied to any equipment or system level, at any level of maintenance, and for any maintenance concept pertinent to avionics, ground electronics, and shipboard electronics. (While the prediction methods were developed specifically for electronic equipments and systems, there is nothing inherent in the methods that should prevent them from being applicable to electro-mechanical or mechanical equipments or systems).

1.3 Basic parameters of measure. Mean time to repair (MTTR) is the primary maintainability parameter that can be predicted using this procedure. The other maintainability parameters that can be predicted using this procedure are: maximum corrective maintenance time at the Φ percentile ($M_{\max}(\Phi)$), percent of faults isolatable to a single replaceable item (I_1); percent of faults isolatable to $<N$ replaceable items (I_N), mean maintenance labor hours per repair (MLH/repair), mean

² A replaceable item (RI) is any of those physical entities normally removed and replaced to effect repair at the maintenance level for which the prediction is being made.

maintenance labor hours per operating hour (MLH/OH), mean maintenance labor hours per flight hour (MLH/FH). (For details see paragraph 2.2).

1.4 Information required. These data items must be provided as part of the maintainability prediction.

1.4.1 Method A. To use Method A the following data are necessary:

- a. The number and contents of (either actual or estimated) the primary RIs
- b. The failure rates, either predicted or estimated, associated with each RI
- c. The basic fault isolation test strategy of each RI
- d. The replacement concept, if fault isolation is to a group of RIs
- e. The packaging philosophy
- f. The fault isolation resolution, either estimated or required (i.e., % of faults isolated to one repairable item or the average RI group size)

1.4.2 Method B. The data necessary to implement Method B are:

- a. The replacement concept of each RI or group of RIs
- b. The fault detection and isolation outputs associated with each RI
- c. The failure rate of each RI
- d. The maintenance procedure that is followed to remove and replace each RI

2. BASIC DEFINITIONS AND MODELS

2.1 MTTR elements. Corrective maintenance (CM) actions consist of the following tasks: Preparation, Fault Isolation, and Fault Correction (further broken down into Disassembly, Interchange, Reassembly, Alignment, and Checkout). The time to perform each of these tasks is an element of MTTR, so the task times are called MTTR elements.

The definitions and abbreviations for the MTTR elements used in the prediction models are shown in Table D-1.

Table D-1. MTTR Elements

<u>MTTR ELEMENT (Abbreviation)</u>	<u>Definition</u>
Preparation (TP_{nj})	Time associated with those tasks required to be performed before fault isolation can be executed.
Fault Isolation (TFI_{nj})	Time associated with those tasks required to isolate the fault to the level at which fault correction begins.
Disassembly (TD_{nj})	Time associated with gaining access to the replaceable item or items identified during the fault isolation process.
Interchange (TI_{nj})	Time associated with the removal and replacement of a faulty replaceable item or suspected faulty item.
Reassembly (TR_{nj})	Time associated with closing up the equipment after interchange is performed.
Alignment (TA_{nj})	Time associated with aligning the system or replaceable item after a fault has been corrected.
Checkout (TC_{nj})	Time associated with the verification that a fault has been corrected and the system is operational.
Start-Up ($TS_{T_{nj}}$)	Time associated with bringing a system up to the operational state it was in prior to failure, once a fault has been corrected and the operational status of the system verified.

The n_j subscript indicates that the n^{th} RI is the object of the maintenance action brought about by the j^{th} fault detection and isolation (FD&I) indication or symptom. The term FD&I is defined as those indications, symptoms, printouts, readouts, or the results of manual procedures which separately, or in combination, indicate a fault or failure has occurred and identifies to the maintainer the procedures to follow in performing maintenance.

Table D-2 indicates the MTTR elements that must be predicted for each of the different fault isolation cases listed at the top of the table.

2.2 Basic models. The maintainability prediction contains two separate procedures: (1) Method A is an early procedure for implementation when preliminary design data are available. (2) Method B is a detailed procedure for implementation when detailed design and support data are available. Both of these procedures are time synthesis model techniques and employ the same general MTTR prediction model.

Table D-2. MTTR Elements for Prediction Procedure

Isolation to Single RI	Isolation to Group with Group Replacement	Isolation to Group with Iterative Replacement	Isolation with Ambiguity (Requires Further Isolation)	
Preparation	Preparation	Preparation	Preparation	Secondary Preparation
Isolation	Isolation	Isolation	Isolation	Secondary Isolation
Disassembly	Disassembly	Disassembly	Disassembly	Secondary Fault Correction
Interchange	Interchange	Interchange	Interchange	
Reassembly	Reassembly	Reassembly	Reassembly	
Alignment	Alignment	Alignment	Alignment	
Checkout	Checkout	Checkout	Checkout	Start Up
Start Up	Start Up	Start Up	Continue	

2.2.1 Mean time to repair (MTTR).

$$MTTR = \frac{\sum_{n=1}^N \lambda_n R_n}{\sum_{n=1}^N \lambda_n}$$

where:

N = number of replaceable items (RI)

λ_n = failure rate of the n^{th} RI

R_n = mean repair time of the n^{th} RI as computed in 2.2.1.1

2.2.1.1 Mean repair time for the n^{th} RI.

$$R_n = \frac{\sum_{j=1}^J \lambda_{nj} \cdot R_{nj}}{\sum_{j=1}^J \lambda_{nj}}$$

where:

J = number of unique FD&I outputs (see 3.2.3)

λ_{nj} = failure rate of those n^{th} RI parts which would cause the n^{th} RI to be called out in the j^{th} FD&I output.

R_{nj} = average repair time of the n^{th} RI when called out in the j^{th} FD&I output as computed in 2.2.1.2

2.2.1.2 Average repair time for the nth RI.

$$R_{nj} = \sum_{m=1}^{M_{nj}} T_{M_{nj}}$$

where:

M_{nj} = # of steps to perform CM when a failure occurs in the nth RI and results in the jth FD&I outputs.
Includes all CM tasks, including operations on other RIs called out in the jth fault isolation result.

$T_{m_{nj}}$ = Average time to perform the mth CM step for the nth RI given the jth FD&I output.

2.2.2 Percent isolation to a single RI.:

$$I_1 = \frac{\sum_{k=1}^K \lambda_k}{\sum_{n=1}^N \sum_{j=1}^J \lambda_{nj}} \times 100$$

where:

λ_{nj} = failure rate of those nth RI parts which would cause the nth RI to be called out in the jth FD&I output.

λ_k = failure rate associated with the kth FD&I output which results in isolation to one RI.

K = number of FD&I outputs which result in isolation to a single RI.

J = number of unique FD&I outputs (see 3.2.3)

2.2.3 Percent isolation to a group of RIs.

$$I_N = \frac{\sum_{p=1}^P \lambda_p}{\sum_{n=1}^N \sum_{j=1}^J \lambda_{nj}} \times 100$$

where:

λ_p = failure rate associated with the pth FD&I output which results in isolation to N or less RIs.

λ_{nj} = same as for I_1

P = number of FD&I outputs which result in isolation to N or less RIs.

Other maintenance parameters that can be predicted using these procedures follow.

2.3 Mean maintenance labor hours per repair (MLH/repair).

$$\overline{\text{MLH}}/\text{Repair} = \frac{\sum_{n=1}^N \lambda_n \overline{\text{MLH}}_n}{\sum_{n=1}^N \lambda_n}$$

where:

N = quantity of RI's
 λ_n = failure rate of nth RI
 $\overline{\text{MLH}}_n$ = mean maintenance labor hours required to repair the nth RI

2.3.1 Mean maintenance laborhours required to repair the nth RI

$$\overline{\text{MLH}}_n = \frac{\sum_{j=1}^J \lambda_{nj} \text{MLH}_{nj}}{\sum_{j=1}^J \lambda_{nj}}$$

where:

J = quantity of FD&I results
 λ_{nj} = failure rate associated with the jth result for the nth RI
 MLH_{nj} = maintenance labor hours required to repair the nth RI given the jth FD&I result

(Method A and Method B procedures can be used by replacing repair times in the appropriate method with the maintenance labor hours required for each repair action.)

2.4 Mean maintenance labor hours per maintenance action (MLH/MA).

This calculation is the same as for MLH repair except that time spent as a result of system failure false alarms must also be included in the maintenance labor hours.

Two types of false alarms are considered:

- 1) Type 1 false alarm is detected during normal operations but cannot be repeated during the fault isolation process.
- 2) Type 2 false alarm is detected and isolated to an RI when the RI does not have an actual fault.

$$\overline{MLH}/MA = \frac{\sum_{n=1}^N (1 + F_{2n}) \lambda_n \overline{MLH}_n + \sum_{n=1}^N F_{1n} \lambda_n \overline{MLH}_D}{\sum_{n=1}^N (1 + F_{2n}) \lambda_n + \sum_{n=1}^N F_{1n} \lambda_n}$$

where:

F_{1n}	=	frequency of occurrence of type 1 false alarms $\underline{1/}$
F_{2n}	=	frequency of occurrence of type 2 false alarms $\underline{1/}$
		$\underline{1/}$ is expressed as a fraction of the n^{th} RI failure rate
$\overline{MLH}_D$	=	mean maintenance labor hours associated with Type 1 false alarms.
$\overline{MLH}_n$	=	mean maintenance labor hours required to repair the n^{th} RI
λ_n	=	failure rate associated with the j^{th} result for the n^{th} RI

2.5 False alarm rates. False alarms are dependent on the system type, operating environment, maintenance environment, system design and fault detection and isolation implementation. Therefore, a standard set of false alarm values would be impossible to derive.

2.6 Mean maintenance labor hours per operating hour (MLH/OH).

MLH/OH includes the entire labor power that is required to maintain a system; corrective maintenance, preventive maintenance, and maintenance caused by false alarms.

$$\overline{MLH}/OH = \sum_{n=1}^N (1 + F_{2n}) \cdot \lambda'_n \cdot \overline{MLH}_n + \sum_{n=1}^N F_{1n} \cdot \lambda'_n \cdot \overline{MLH}_D + \sum_{r=1}^{PM} F_r \cdot \overline{MLH}_r$$

where:

λ'_n	=	λ_n expressed in failures per operating hour
F_r	=	frequency of r^{th} preventive maintenance
$\overline{MLH}_r$	=	mean maintenance labor hours to perform r^{th} preventive maintenance type
PM	=	quantity of unique preventive maintenance types
N	=	quantity of RI's
F_{1n}	=	frequency occurrence of type 1 false alarms $\underline{1/}$
F_{2n}	=	frequency of occurrence of type 2 false alarms $\underline{1/}$
$\overline{MLH}_D$	=	mean maintenance labor hours associated with type 1 false alarms
$\overline{MLH}_n$	=	mean maintenance labor hours required to repair the n^{th} RI

$\underline{1/}$ expressed as a fraction of the n^{th} RI failure rate.

2.7 Mean maintenance labor hours per flight hour (MLH/FH). MLH/FH is the same as $\overline{MLH}/OH$ where $\lambda'_n = \lambda_n$ is expressed in failures per flight hour.

2.8 Maximum corrective maintenance time for the percentile ($M_{\max}(\Phi)$). Two $M_{\max}(\Phi)$ models are provided. The first yields an approximate value and requires that system repair times be log-normally distributed. The second gives a more accurate value.

2.8.1 Approximate $M_{\max}(\Phi)$. Appendix B of MIL-HDBK-472 contains tables of $M_{\max}(\Phi)$ values for selected values of system Φ , system MTTR (MEAN), and standard deviation of system repair times (SIGMA). MTTR may be predicted using Method A, and the MTTR models in paragraph 2.2.1. SIGMA is usually determined from data on similar equipments. Approximate $M_{\max}(\Phi)$ values for values of Φ , MEAN, and SIGMA not covered in the Appendix B of MIL-HDBK-472 may be calculated by using the following equation.

$$M_{\max}(\Phi) = \exp \left[\log \text{MTTR} + F \text{SIGMA} \right]$$

where:

$$\text{SIGMA} = \sqrt{\frac{\sum_{i=1}^N (\log R_{n_i})^2 - \left[\sum_{i=1}^N (\log R_{n_i}) \right]^2 / N}{N - 1}}$$

2.8.2 Accurate $M_{\max}(\Phi)$. A computerized and a manual means of predicting an accurate $M_{\max}(\Phi)$ are provided in Appendixes C and D, respectively, of MIL-HDBK-472.

3. APPLICATION. The application of the early and detailed maintainability prediction techniques is described in 3.1 and 3.2 respectively.

3.1 Method A - early prediction procedure. This section provides a step-by-step procedure for performing an early prediction of mean time to repair. The tasks involved in performing the early prediction are:

- a. Define the prediction requirements
- b. Define the replacement concept
- c. Determine the prediction parameters
- d. Select the appropriate models
- e. Compute the MTTR (or other parameters)

Descriptions of each of these tasks are provided in the following subsections.

3.1.1 Prediction requirement definition. This step of the prediction is in some respects the most important aspect since it establishes a common baseline of understanding the prediction purpose, approach and scope. During this step, the maintainability parameter(s) to be evaluated is defined, the prediction ground rules are established, and the maintenance level for which the prediction is being made is defined.

Parameter definition includes the selection (if required) of the parameter(s) to be evaluated and the establishment of a qualitative and quantitative definition of each parameter. If the prediction is being performed in compliance with a customer statement of work defining the parameter to be analyzed, it must be determined if the stated parameter is consistent with an equivalent parameter contained in this methodology. If not, the prediction models must be changed accordingly. As part of the parameter evaluation, it must be determined which elemental maintenance tasks (e.g., preparation, isolation, etc.) are to be included in the analysis and which are to be excluded.

The latest aspect of this step is to explicitly define the maintenance level for which the prediction is being made. If the level is defined in terms of a specific maintenance organization (e.g., direct support unit, depot, etc.), then the tasks to be performed are readily defined by the maintenance concept as described in the following section. If the level is defined by operating level or location (e.g., on-site, flight-line, etc.), then this level must be redefined in terms of the maintenance organization(s) performing maintenance at the level/location.

3.1.2 Replacement concept definition. The maintenance concept must be established, so that in conjunction with a definition of the prediction requirements (paragraph 3.1.1), a baseline is established which defines the prediction to be performed. With respect to the maintainability prediction, the primary output of the maintenance concept is the definition of how a repair is effected and what the replaceable items are.

As part of this process, a complete set of replaceable items is identified. If the maintenance concept allows for fault isolation to a group of RIs and repair by group replacement, then the RI groups can be reclassified as RIs if each of the groups is independent of other groups.

3.1.3 Determination of the prediction parameters. This step involves:

- a. Defining the RIs
- b. Determining the predicted or estimated failure rate associated with each RI
- c. Defining fault isolation test methodology for each RI
- d. Defining the replacement concept
- e. Defining the packaging philosophy
- f. Determining the estimated or required fault isolation resolution (i.e., X% to 1 RI or average RI group size).

Forms similar to those in Figures D-1 and D-2 should be used for the data collection process. Data are collected on these forms at the level for which predictions are performed. For example, if a repair time is to be computed for every equipment within a system, then a separate data collection form should be used for each equipment. Data should be tabulated in the following manner.

- a. First tabulate all the primary RIs and their associated failure rates in the respective columns of Figure D-1 (V refers to the method).
- b. Next describe all methods (V) for performing each elemental activity (m) in Figure D-2. (Note: some maintenance actions do not include all maintenance elements).
- c. Next enter the appropriate number of headings (V_m) for each elemental activity along the top of Figure D-1.

$$MTTR = \bar{T}_p + \bar{T}_{FI} + \bar{T}_{FC} + \bar{T}_A + \bar{T}_{co} + \bar{T}_{ST} = \sum_{m=1}^M \bar{T}_m$$

where:

$\bar{T}_p$	=	Average Preparation Time
$\bar{T}_{FI}$	=	Average Fault Isolation Time
$\bar{T}_D$	=	Average Disassembly Time
$\bar{T}_I$	=	Average Interchange Time
$\bar{T}_R$	=	Average Reassembly Time
$\bar{T}_A$	=	Average Alignment Time
$\bar{T}_{CO}$	=	Average Checkout Time
$\bar{T}_{ST}$	=	Average Startup Time
$\bar{T}_{FC}$	=	$\bar{T}_D + \bar{T}_I + \bar{T}_R$
$\bar{T}_m$	=	Average time of the m^{th} element of MTTR

Variations of the model are limited to deleting the time elements for elemental activity terms that are not necessary to complete certain maintenance actions.

The selection of submodels is dependent on the replacement policy imposed. The appropriate submodels for computing the average time for the elemental activities and the MTTR submodel term definitions are given in Figures D-3 and D-4, respectively.

Maintenance Philosophy	Preparation T_P	Fault Isolation T_{FI}	Disassembly/Reassembly T_D/T_R	Interchange T_I	Alignment T_A	Check Out T_C	Startup T_{ST}
Isolation on a Single RI	↑	↑	$\bar{T}_m = \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$	$\bar{T}_I = \frac{V_I \sum_{v=1}^I \lambda_{Iv} T_{Iv}}{\sum_{v=1}^I \lambda_{Iv}}$	↑		↑
Isolation to a Group of RIs	Group Replacement	Multiple Access	$\bar{T}_m = \sum_{a=1}^A \left[1 - \left(\frac{\lambda_T - \lambda_a}{\lambda_T} \right)^{S_G} \right] T_{ma}$ $m = D, R$	$\bar{T}_I = (\bar{S}_G) \frac{V_I \sum_{v=1}^I \lambda_{Iv} T_{Iv}}{\sum_{v=1}^I \lambda_{Iv}}$		$\bar{T}_C = \frac{V_C \sum_{v=1}^C \lambda_{Cv} T_{Cv}}{\sum_{v=1}^C \lambda_{Cv}}$	
		Single Access	$m = \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$				
	Reactive Replacement	Multiple Access	$\bar{T}_P = \frac{V_P \sum_{v=1}^P \lambda_{Pv} T_{Pv}}{\sum_{v=1}^P \lambda_{Pv}}$ $\bar{T}_{FI} = \frac{V_{FI} \sum_{v=1}^{V_{FI}} \lambda_{FIv} T_{FIv}}{\sum_{v=1}^{V_{FI}} \lambda_{FIv}}$ $\bar{T}_m = \left(\frac{A+1}{2} \right) \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$	$\bar{T}_I = \frac{V_I \sum_{v=1}^I \lambda_{Iv} T_{Iv}}{\sum_{v=1}^I \lambda_{Iv}}$	$\bar{T}_{IA} = \frac{V_A \sum_{v=1}^A \lambda_{Av} T_{Av}}{\sum_{v=1}^A \lambda_{Av}}$	$\bar{T}_{ST} = \frac{V_{ST} \sum_{v=1}^{V_{ST}} \lambda_{STv} T_{STv}}{\sum_{v=1}^{V_{ST}} \lambda_{STv}}$	
		Reassembly Not Required for Checkout					
		Reassembly Required for Checkout	$\bar{T}_m = (\bar{S}_I) \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$	$\bar{T}_I = (\bar{S}_I) \frac{V_I \sum_{v=1}^I \lambda_{Iv} T_{Iv}}{\sum_{v=1}^I \lambda_{Iv}}$		$\bar{T}_C = (\bar{S}_I) \frac{V_C \sum_{v=1}^C \lambda_{Cv} T_{Cv}}{\sum_{v=1}^C \lambda_{Cv}}$	
		Reassembly Not Required for Checkout	$\bar{T}_m = \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$				
		Reassembly Required for Checkout	$\bar{T}_m = (\bar{S}_I) \frac{V_m \sum_{v=1}^m \lambda_{mv} T_{mv}}{\sum_{v=1}^m \lambda_{mv}}$ $m = D, R$				

Figure D-3. MTTR submodels.

T_{Pv}	time required to prepare a system for fault isolation using the v^{th} method
T_{Fiv}	time required to isolate a fault using the v^{th} method
T_{Dv}	time required to perform disassembly using the v^{th} method
T_{Rv}	time required to perform reassembly using the v^{th} method
T_{Iv}	time required to interchange an RI using the v^{th} method
T_{Av}	time required to align or calibrate an RI using the v^{th} method
T_{Cv}	time required to check a repair using the v^{th} method
T_{STv}	time required to start up a system using the v^{th} method
λ_{Pv}	failure rate of RIs associated with the v^{th} method of performing preparation
λ_{Fiv}	failure rate of RIs associated with the v^{th} method of performing fault isolation
λ_{Dv}	failure rate of RIs associated with the v^{th} method of performing disassembly
λ_{Rv}	failure rate of RIs associated with the v^{th} method of performing reassembly
λ_{Iv}	failure rate of RIs associated with the v^{th} method of performing interchange
λ_{Av}	failure rate of RIs associated with the v^{th} method of performing alignment
λ_{Cv}	failure rate of RIs associated with the v^{th} method of performing checkout
λ_{STv}	failure rate of RIs associated with the v^{th} method of performing set-up
V_P	the number of unique ways to perform preparation
V_{FI}	the number of unique ways to perform fault isolation
V_D	the number of unique ways to perform disassembly
V_R	the number of unique ways to perform reassembly
V_I	the number of unique ways to perform interchange
V_A	the number of unique ways to perform alignment
V_C	the number of unique ways to perform check-out
V_{ST}	the number of unique ways to perform start-up
$\overline{S}_G$	the average number of RIs contained in a fault isolation result
$\overline{S}_I$	the average number of interchanges required to correct a fault
A	the number of unique accesses ($A - V_D$ or V_R)
$\overline{A}$	the average number of unique accesses required per fault isolation result
λ_a	the failure rate of the RIs that require the a^{th} type of access
λ_T	the total system failure rate
T_{Da}	the time required to disassemble the a^{th} access
T_{Ra}	the time required to reassemble the a^{th} access

Figure D-4. Definitions of MTTR submodel terms.

The submodels presented are of a general form and can generally be applied to any equipment level (i.e., system, subsystem, equipment, etc.). The only limitation being that if the fault isolation result ($\bar{S}_G$) or the average number of iterations required to correct a fault ($\bar{S}_I$) are computed, the prediction level must be consistent with the RI grouping rules presented in paragraph 3.1.5.1. Otherwise, the elemental activity submodels are applied at the lowest level for which an MTTR prediction is desired.

3.1.5 MTTR computation. The MTTR is computed at the level at which $\bar{S}_G$ or $\bar{S}_I$ is established. For example, if $\bar{S}_G$ or $\bar{S}_I$ can be estimated for each equipment within a system, then the lowest level that the MTTR can be predicted is the equipment level. Higher level predictions of MTTR, such as system level MTTR, can be computed by taking a failure rate weighted average of the equipment MTTRs within the system.

Computing repair times below the level at which $\bar{S}_I$ or $\bar{S}_G$ is established may give an inaccurate account of repair times. The only exception is when fault isolation is down to a single RI ($\bar{S}_G = 1$) for the entire system, equipment . . .), then the MTTR may be computed at any level since ambiguities between RIs do not exist. Otherwise, the following criteria must be followed:

In order to compute a repair time at a given level, a value for $\bar{S}_I$ or $\bar{S}_G$ must be established at that level. After the level at which the repair times will be computed has been selected, the appropriate models are selected to compute time for each elemental activity at that level with higher level repair time being computed using a failure rate weighted average.

Values for $\bar{S}_G$, $\bar{S}_I$, $\bar{A}$, $\bar{T}_D$ or $\bar{T}_R$, where required, should be computed as detailed in the following subsections.

3.1.5.1 Method of computing $\bar{S}_G$ and $\bar{S}_I$. Two methods are used for computing the average number of RIs in a fault isolation result ($\bar{S}_G$) and the average number of iterations required to correct a fault ($\bar{S}_I$), compute $\bar{S}_I$ (I or G) using the specified or design requirements, or compute $\bar{S}_I$ (I or G) by assessing the approximate fault isolation capabilities of the system.

3.1.5.1.1 Method 1. The first method of computing $\bar{S}_I$ or $\bar{S}_G$ depends upon how the fault isolation requirements are specified. In the fault isolation resolution is specified as follows:

$X_1\%$ to $\leq N_1$ RIs

N_1 RIs < $X_2\%$ to $\leq N_2$ RIs

N_2 RIs $\leq X_3\%$ to $\leq N_3$ RIs

and $X_1 + X_2 + X_3 = 100$

then,

$$\bar{S}_{I \text{ or } G} = \frac{X_1\left(\frac{N_1 + 1}{2}\right) + X_2\left(\frac{N_1 + N_2 + 1}{2}\right) + X_3\left(\frac{N_2 + N_3 + 1}{2}\right)}{100}$$

If the fault isolation requirements are specified as follows:

$X_1\%$ to $\leq N_1$ RIs

$X_2\%$ to $\leq N_2$ RIs

100% to $\leq N_3$ RIs

where:

$$X_1\% < X_2\% < 100\%$$

then,

$$\bar{S}_{I \text{ or } G} = \frac{X_1 \left(\frac{N_1 + 1}{2} \right) + (X_2 - X_1) \left(\frac{N_1 + N_2 + 1}{2} \right) + (100 - X_2) \left(\frac{N_2 + N_3 + 1}{2} \right)}{100}$$

The predicted MTTR using this method of computing S is based on the assumption that the specified fault isolation requirements have been (or will be) met. The resulting prediction is the inherent MTTR that will be realized by achieving the specified requirements. This approach is valuable during the early stages of equipment development for purposes of allocation and assessment of the requirements facility. This approach should not be used when data are available on the actual fault isolation characteristics.

3.1.5.1.2 Method 2. The second method of computing $\bar{S}_I$ or $\bar{S}_G$ involves an analysis of the fault isolation characteristics of the subject equipment/system as follows:

- a. Prepare a simple block diagram depicting the system and how each major function is related (i.e., show functional interfaces).
- b. Group the functions (RIs) into "G" RI sets such that:
 - an estimate of the fault isolation (number of RIs) can be determined for each RI set
 - each RI set is independent of any other RI set
 - each RI set established is the smallest set that can be established
- c. For each RI set (g) estimate the average fault isolation resolution or the average number of RIs per fault isolation result depending on the replacement philosophy in question ($\bar{S}_{(I)g}$ if iterative replacement, $S_{(G)g}$ if group replacement).
- d. Compute the average $\bar{S}_I$, or $\bar{S}_G$ for the system or equipment using a failure rate weighted model.

$$\bar{S}_I \text{ or } \bar{S}_G = \frac{\sum_{g=1}^G \lambda_g \bar{S}_g}{\sum_{g=1}^G \lambda_g}$$

For repair times computed at lower levels, the overall $\bar{S}$ does not have to be computed.

3.1.5.2 The average number of accesses (disassemblies and reassemblies) required per fault isolation result ($\bar{A}$) can be computed as follows.

$$\bar{A} = \frac{\sum_{g=1}^G \lambda_g \bar{A}_g}{\sum_{g=1}^G \lambda_g}$$

and,

$$\bar{A}_g = \sum_{a=1}^{A_g} P_{ga} = \sum_{a=1}^{A_g} \left[1 - \frac{\lambda_g - \lambda_{ga} \bar{S}_g}{\lambda_g} \right]$$

where:

- $\bar{A}_g$ = average number of accesses required per fault isolation result in g^{th} RI set, ("G" RI sets established the same way as was done for S)
- P_{ga} = the probability that the a^{th} access will be required for any random fault isolation result
- A_g = the number of unique accesses in the g^{th} RI set
- λ_g = the failure rate of the RIs located in the g^{th} RI set
- λ_{ga} = failure rate of the RIs located in a^{th} access location of g^{th} RI set
- $\bar{S}_g$ = average number of RIs per fault isolation result for the g^{th} RI set
- G = total number of RI sets

The computation of $\bar{T}_{D'}$ and $\bar{T}_{R'}$ is exactly like the method used for $\bar{A}$ with one modification. Each probability is multiplied by its appropriate disassembly or reassembly time. The equation for $\bar{T}_{D'}$ or $\bar{T}_{R'}$ is:

$$\bar{T}_{D'} = \frac{\sum_{g=1}^G \lambda_g \bar{T}_{Dg'}}{\sum_{g=1}^G \lambda_g}$$

and,

$$\bar{T}_{R'} = \frac{\sum_{g=1}^G \lambda_g \bar{T}_{Rg'}}{\sum_{g=1}^G \lambda_g}$$

where:

$$\bar{T}_{Dg'} = \sum_{a=1}^{A_g} \left[1 - \frac{\lambda_g - \lambda_{ga} \bar{S}_g}{\lambda_g} \right] T_{Dga}$$

The same equation also hold true for reassembly, ($\bar{T}_{Rg'}$)

where:

- T_{Dga} = the disassembly time for the a^{th} access of the g^{th} RI set.
- T_{Rga} = the reassembly time for the a^{th} access of the g^{th} RI set.

Note here also that if the RIs are grouped into just one set instead of G sets, then all the subscripts "g" will fall-out and the failure rate weighting of the gth RI sets is not necessary.

3.1.5.3 Determination of MTTR. The MTTR can now be computed by summing up the average times computed from each submodel. Thus, the MTTR is expressed as

$$MTTR = \sum_{m=1}^M \bar{T}_m$$

If the repair time computed is for a lower level, then the higher level repair times are computed as follows:

$$MTTR = \frac{\sum_{b=1}^B \lambda_b MTTR_b}{\sum_{b=1}^B \lambda_b}$$

where:

$MTTR_b$	=	mean repair time of the b th lower level
λ_b	=	failure rate of the b th lower level
B	=	quantity of lower level breakdowns.

3.2 Method B - Detailed prediction procedure. This section provides a step by step procedure for performing a detailed prediction of mean time to repair (MTTR). The tasks involved in performing the prediction are:

- a. Define the prediction requirements
- b. Define the replacement concept
- c. Identify the fault detection and isolation outputs (FD&I outputs)
- d. Correlate the FD&I outputs and hardware features
- e. Correlate replaceable items and fault detection and isolation outputs
- f. Prepare a maintenance flow diagram
- g. Prepare time line analyses
- h. Compute the maintainability parameters

Descriptions of each of the tasks are provided in the following subsections.

3.2.1 Prediction requirements definition. This step is similar to that required for an early prediction; refer to paragraph 3.1.1.

3.2.2 Replacement concept definition. This step is similar to that required for an early prediction; refer to paragraph 3.1.2.

3.2.3 Fault detection and isolation output identification. This step involves the identification of all the "outputs" which are used in the fault detection and isolation process. Normally, the fault detection and isolation processes are segregated. However, for purposes of maintainability prediction, the fault detection methodology is considered as the first step of fault isolation and is properly included as part of the isolation capability. Any time associated with fault detection (e.g., mean fault detection time) is normally excluded from the prediction model, but can be included if desired.

The term fault detection and isolation outputs is defined as those indications, symptoms, print-outs, readouts, or the results of manual procedures which separately or in combination, identify to the maintenance technician the procedure to be followed.

FD&I outputs will vary in form, format, complexity and data content from system to system and some will be more obvious than others. The maintenance actions taken in response to these outputs may depend upon the system maintenance environment and the system operating criticality. It is important, therefore, not only to identify the FD&I outputs but also to ensure that the FD&I outputs identified are the ones that will be used in the intended maintenance environment.

Some of the more common generic FD&I outputs are:

- a. Indicator or annunciator
- b. Diagnostic or BIT output
- c. Meter readings
- d. Circuit breaker and fuse indicators

- e. Display presentation
- f. Alarms
- g. Improper system operation
- h. Improper system response
- i. System operating alerts

To apply the prediction methodology presented herein, the predictor should first identify all primary unique outputs upon which the maintenance technician relies to make decisions on the repair methodology (e.g., perform adjustment, replace RI, proceed to a different method of fault isolation, etc.). Secondary outputs should then be identified for those cases where the primary output yielded a result which did not correct the problem and further isolation is required.

3.2.4 FD&I outputs and hardware correlation. The key to this prediction procedure, and by far the most demanding of the prediction tasks, is the establishment of a correlation between the FD&I outputs (See paragraph 3.2.7) and the hardware for which the prediction is being made. This step demands a thorough understanding of the system hardware, software, monitoring and diagnostic capabilities, and of the FD&I features inherent to the system. FD&I features are those hardware and software elements, or combinations thereof, which generate or cause to be generated each FD&I output.

This task can be accomplished either from the top down or bottom up. The top down approach involves a fault tree technique where the top of the tree is each unique FD&I output; the next tier identifies the FD&I feature(s) which can yield the subject output; and, the bottom tier identifies the RIs or partial RIs which upon failure would be detected or isolated by the subject FD&I feature. The bottom up approach involves identification of all the circuitry in terms of RIs associated with each FD&I feature, and the analysis of how a failure of each RI presents itself in terms of an FD&I output.

Either approach requires the same five steps to be performed:

- a. Identify all FD&I features
- b. Identify the circuitry associated with each feature
- c. Identify the FD&I sequencing
- d. Establish the RI failure rate associated with each FD&I feature
- e. Correlate the FD&I features with the FD&I outputs

FD&I features are those hardware and software elements, or combinations thereof, which generate or cause to be generated each FD&I output. Typical features include diagnostic program routines, BIT routines, BITE, performance monitoring programs, status monitors, and test points.

After the FD&I features are identified, the circuit schematics are analyzed to identify the components tested or verified by each feature. The outputs of this analysis are then translated into a matrix as shown in Figure D-5. The matrix identifies, for each FD&I feature, the RIs and components which are tested by that feature. Also included in the matrix is an identifier which defines the order in which the FD&I features are utilized during the isolation process.

Figure D-5: Matrix for correlating FD&I features with RIs

Nomenclature		Fault Detection and Isolation Features						
RI/Component	Failure Rate	Feature 1 Seq No.	Feature 2 Seq No.	Feature 3 Seq No.	Feature 4 Seq No.	Feature 5 Seq No.	Feature N Seq No.	Manual Isolation Failure Rate
RI No. 1								
Component A								
Component B								
Component C								
Component D								
Component E								
Component F-M								
Component N								
Component P-V								
RI No. 2								
Component A								
Component B								

*Component - Same lower level of assembly

The matrix is used to identify the failure rate of each RI associated with each FD&I feature. The first FD&I feature is examined and the failure rate of each component associated with that feature is entered in the matrix under that feature. The second feature is then examined, etc. If a component is tested by more than one feature, the failure rate is assigned to the first feature which would result in a positive failure indication. If different tests of the same component check different failure modes, then the failure rate is apportioned to each feature based on the relative occurrence of each failure mode. The failure rates for the components under each RI in each FD&I feature column are summed and entered as the failure rate for the RI checked by that particular feature. This assumes the feature either checks a single RI or can check multiple RIs by some sequencing scheme. Components not included under any FD&I features represent failures not isolated with the FD&I features. The failure rates of the failures not isolated by the FD&I features are noted in the manual isolation failure rate column of the matrix to complete the accounting of the total equipment failure rate. All manual isolation cases must be accounted for.

In those cases where the n^{th} failure rate is known to result in several FD&I outputs, but the allocated failure rates are not known, the rationale for the assumed allocation of the failure rates shall be stated.

The next step in the correlation process is to associate the FD&I features with the FD&I outputs. This is accomplished using a fault tree type diagram such as the sample shown in Figures D-6 & D-7. The top of the tree consists of all FD&I outputs; the second tier contains the FD&I features which separately or jointly result in the given FD&I output; and, the bottom tier presents the RIs associated with each FD&I feature and the failure rate associated with that feature. The circles are used to assign numbers to all unique FD&I outputs. The triangles identify the order in which RIs are replaced when the replacement concept calls for iterative replacement.

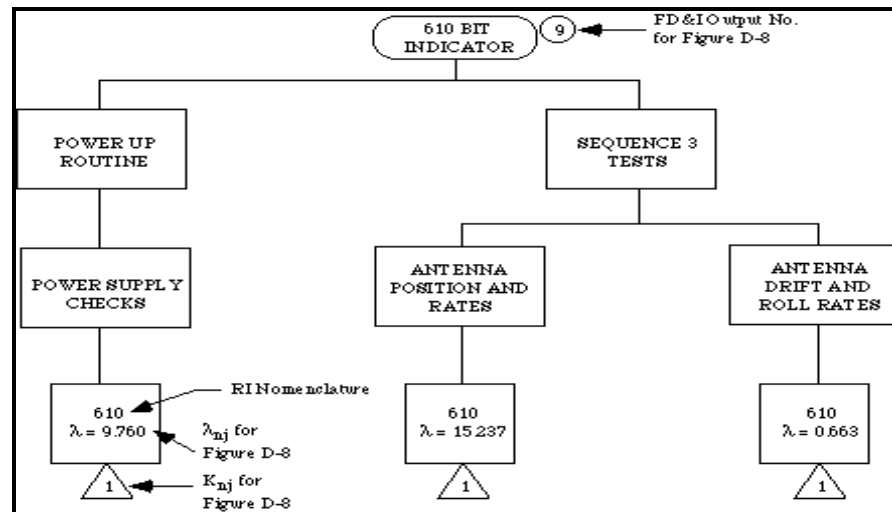


Figure D-6. Fault isolation output and RI correlation tree.

3.2.5 Prepare maintenance flow diagrams. Next a maintenance flow diagram (MFD) is prepared to establish the R_{nj} values for insertion in the Maintenance Correlation Matrix (Figure D-10). The MFD is prepared to illustrate the sequence of maintenance required. The symbols used in the MFD are shown in Table D-3 following Figure D-7.

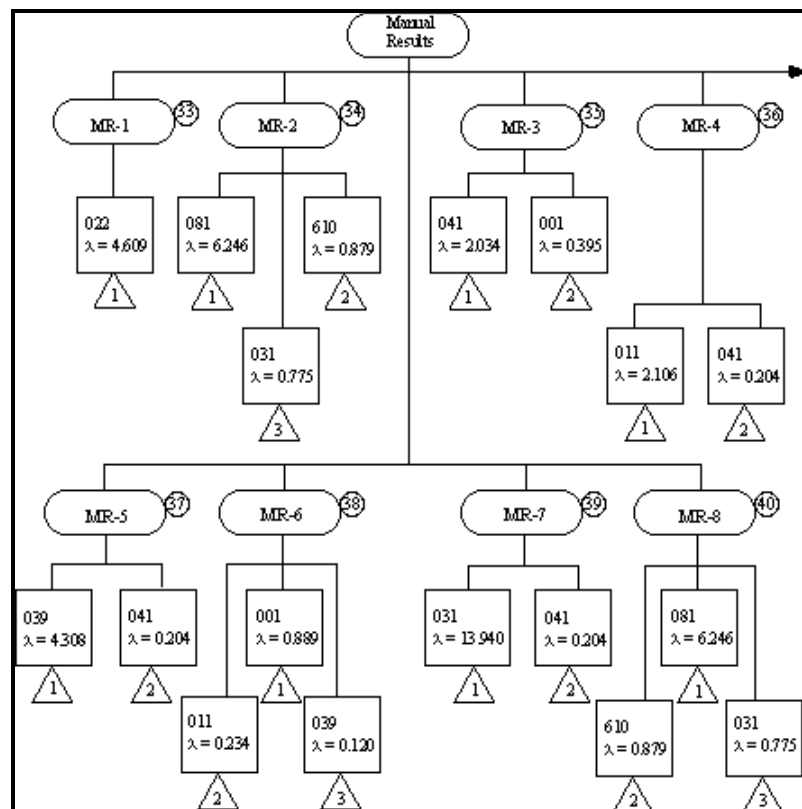


Figure D-7. Manual fault isolation output and RI correlation tree (Partial).

The MFD (as illustrated in Figure D-8) starts on the left side of the figures as a "Failure Occurs and is Detected" event. If isolation is inherent in fault detection, the next item shown in the MFD is the unique FD&I outputs. If isolation is not inherent in detection, the next item in the MFD is the fault detection output. This would be followed by activity blocks which define the procedure

followed to achieve fault isolation. The activity block(s) is followed by the unique primary FD&I outputs associated with the maintenance actions that have been executed.

Following the FD&I output symbols are shown the activities required for fault correction and repair verification.

If a FD&I output results in non-ambiguous maintenance (i.e., primary isolation to a single RI, or group RI replacement), then an "End" symbol will directly follow the fault correction and verification activities. If a FD&I output results in an ambiguous result, a verification decision block is shown after each verification activity (except the last). Any activity (e.g., clean-up) performed after a positive verification decision is shown in an activity block(s) between the decision block and the End symbol. Associated with each End symbol is a path identifier which uniquely identifies each path by RI and FD&I output. For example, the path associated with the second RI and FD&I Output #12 would be designated as 2, 12.

Care must be exercised to ensure that all possible maintenance actions that could be followed as a result of a FD&I output observation have been accounted for, especially those that result in Manual Fault Isolation.

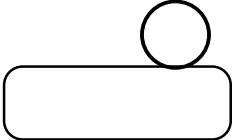
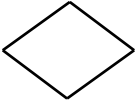
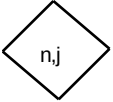
	<u>Starting Point.</u> (i.e., Failure Occurs and is Detected) or <u>Ending Point</u>
	<u>Activity Block.</u> The top of the block indicates a specific maintenance activity and the bottom indicates the time associated with that activity. This is the only symbol that denotes time.
	<u>FD&I Outputs.</u> Designates the primary or secondary unique FD&I output which defines the subsequent maintenance activity to be performed. The "j" associated with the output is entered in the circle.
	<u>Decision Point.</u> Defines a point in the maintenance flow at which time the maintenance technician must make a decision on which subsequent path to take.
	<u>Path Identifier.</u> Uniquely identifies each path by unique RI(n) and FD&I output (j).
	<u>Continuation.</u> Designates continuation from or to another place in the maintenance flow diagram.

Table D-3. Symbols Used in the MFD.

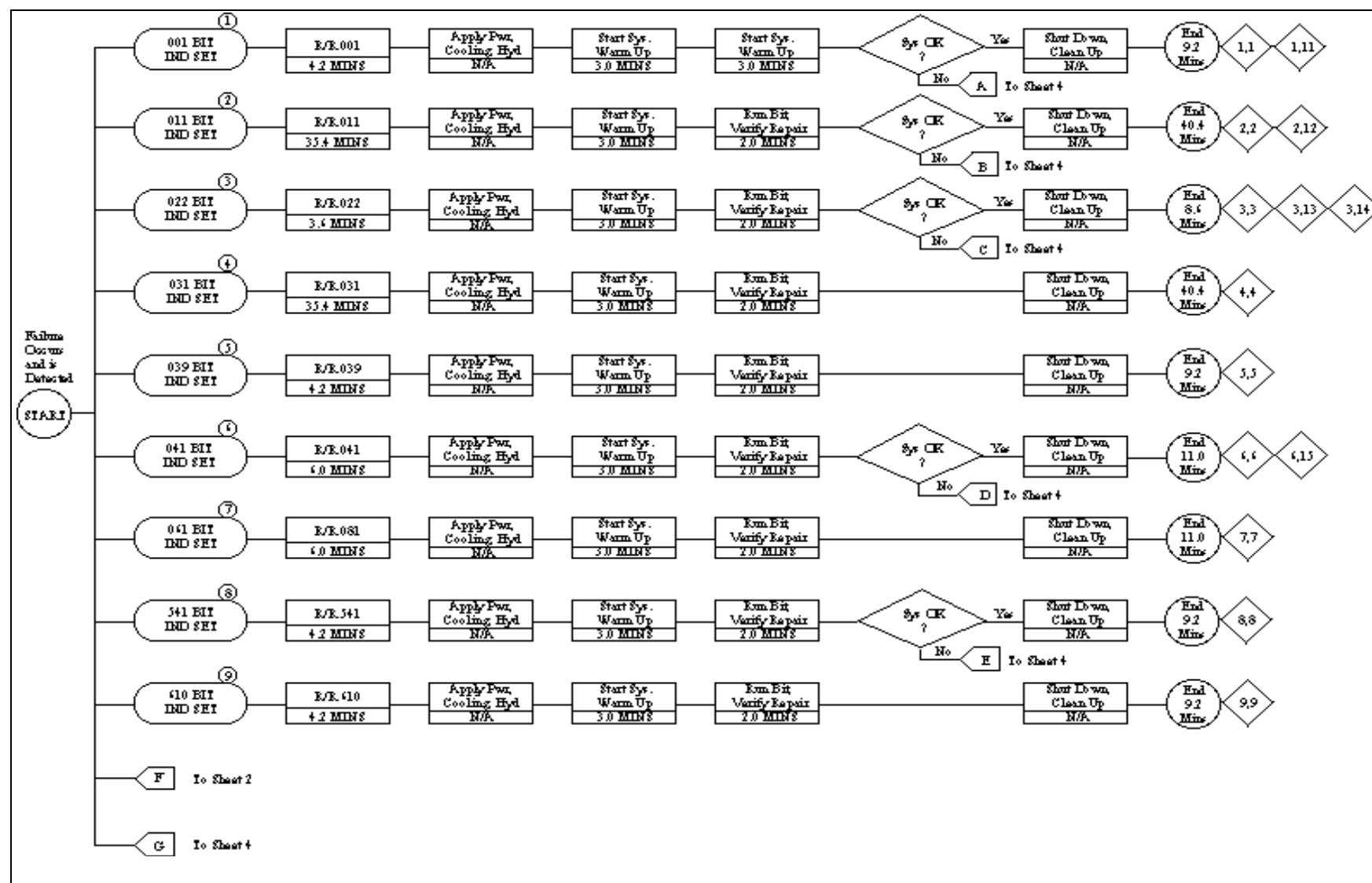


Figure D-8. Sample Maintenance Flow Diagram.

The R_{nj} values are computed by adding the times associated with each activity block from the "Failure Occurs and is Detected" event to the "end" event for the subject (n, j) pair. Note that only the activity blocks have time associated with them. The time entered in the individual activity blocks is computed from a time line analysis prepared in accordance with paragraph 3.2.6. Elemental times entered in the time line analysis are extracted from the following sources in the order given:

- a. Actual times experienced on the subject equipment
- b. Standard times from Tables D-4 and D-5 (see Section 4)
- c. Actual times experienced on similar equipment
- d. Other recognized time sources
- e. Engineering judgment

In the establishment of the time line analysis, the number of maintainers must be considered. For example, if a given equipment has two technicians performing maintenance, one technician may perform disassembly to achieve access to the faulty RI while the second technician simultaneously performs other work. In the maintenance flow diagram, this would show as a single maintenance activity with the associated time being the elapsed clock time. If the parameter of interest was MLH /OH, instead of MTTR, then the time entered in the activity block would be the combined MLH in lieu of the elapsed time.

3.2.6 Time line analysis. The estimated times used in the two prediction methodologies are synthesized using a time line analysis method. A time line analysis consists of computing the total elapsed time of a maintenance action by accounting for the time required to perform each step. The procedure for performing a time line analysis is as follows:

- a. Identify each task that comprises the maintenance action
- b. Determine the time required to perform each task by either actual times, maintenance time standards, time studies, or engineering judgment
- c. Determine which actions can be done simultaneously if more than one maintainer is available
- d. Determine the overall time to perform the maintenance action by summing up the times to perform each action

Figure D-9 is an example of how to synthesize time for a simple physical task. The time associated with each task is extracted from column 3, 4, or 5 of Table D-4.

RI Name: Module T/R Element Maintenance Action: Interchange			
Description of the Elemental Tasks	Time/Action	Qty	Total Time
Remove Quick Release Coax	0.04	4	0.16
Remove Slide Lock Connector	0.09	1	0.09
Remove Module	0.09	1	0.09
Replace Module	0.11	1	0.11
Replace Slide Lock Connector	0.12	1	0.12
Replace Quick Release Coax	0.04	4	0.16
Total Time			0.73

Figure D-9. Example time synthesis analysis.

3.2.7 RI and FD&I output correlation. The results of the preceding section are summarized in a matrix which shows the relationship among the RIs for which the prediction is being performed and the total set of FD&I outputs. The matrix (Figure D-10) identifies the RIs across the top and the unique FD&I Outputs down the left column. In reference to the math models (refer to paragraph 2.2) the RIs are the "n" parameters and the FD&I outputs are the "j" parameters. Each RI column is further divided into three columns: Q_{nj} , λ_{nj} , and R_{nj}

Under each RI column, enter the failure rate (λ_{nj}) of the RI (obtained from the FD&I correlation tree) (See Figure D-6) that is associated with each FD&I output. For each unique output which has only one RI associated with it, enter a 1 in the Q_{nj} column for that combination. For those outputs which are associated with 2 or more RIs, the Q_{nj} value is determined by the replacement concept. If the replacement concept is group RI replacement, enter under Q_{nj} the number of RIs associated with each output. For example, if three RIs could contribute to the same FD&I output, then a 3 is entered in the Q_{nj} for each of those RIs. If the replacement concept is iterative replacement, then Q_{nj} is assigned based on the order of replacement. That is, the first RI to be replaced upon recognition of the subject FD&I output is designated $Q_{nj} = 1$, the second $Q_{nj} = 2$ and so forth. In cases of iterative replacement, the values for each Q_{nj} is based on the relative failure rates of the RIs, with the highest failure rate RI assigned as the first replacement item.

FD&I Outputs (j)	RI n			1			2			3			4			5		
	λ_n			λ_1			λ_2			λ_3			λ_4			λ_5		
	Q_{1j}	λ_{1j}	R_{1j}	Q_{2j}	λ_{2j}	R_{2j}	Q_{3j}	λ_{3j}	R_{3j}	Q_{4j}	λ_{4j}	R_{4j}	Q_{5j}	λ_{5j}	R_{5j}			
1																		
2																		
3																		
4																		
5																		
6																		
-																		
-																		
-																		

Figure D-10. Maintenance correlation matrix format.

3.2.8 Compute maintainability parameters. Once the MFD and Maintenance Correlation Matrix have been completed, compute the maintainability parameter(s) using the equations in Section 2 of this appendix.

4. TIME STANDARDS

The time standards are tabulated in Table D-4. The times tabulated in Table D-4 have corresponding figures referenced which illustrate what each time represents. Table D-5 contains composite times of common maintenance actions that may occur. Columns two and four of Table D-5 denote which times of Table D-4 were used to synthesize each activity (letters denote removal (A) and replaceable (B) times).

It should be noted that the standard times given are ideals. In actual practice, the task times will probably be longer due to environmental conditions, the need for gloves or other protective clothing that may interfere with performing certain tasks, and less than ideal access. For this reason, as stated in 3.2.5, elemental times should be extracted from the following sources in the order given:

- a. Actual times experienced on the subject equipment
- b. Standard times from Tables D-4 and D-5
- c. Actual times experienced on similar equipment
- d. Other recognized time sources
- e. Engineering judgment

4.1 Maintenance task synthesis.

Other maintenance tasks can easily be synthesized by the following method (for an example, see Figure D-9, in paragraph 3.2.6).

- a. List the actions involved for the maintenance task.
- b. Obtain the times for each action by using Table D-4 (times that are not listed should be established either by actual data, time studies, or engineering judgment).
- c. Compute the time by summing up each individual time.

Table D-4. Elemental Maintenance Actions.

Time Standard Number	Description	Standard Times			Reference Fig- ure
		Remove (min.)	Replace (min.)	Interchange (min.)	
FASTENERS					
1	Standard Screws	0.16	0.26	0.42	D-11
2	Hex or Allen Type Screws	0.17	0.43	0.60	D-12
3	Captive Screws	0.15	0.20	0.35	D-13
4	DZUS (1/4 Turnlock)	0.08	0.05	0.13	D-14
5	Tridair Fasteners	0.06	0.06	0.12	D-15
6	Thumbscrews	0.06	0.08	0.14	D-16
7	Machine Screws	0.21	0.46	0.67	D-17
8	Nuts or Bolts	0.34	0.44	0.78	D-18
9	Retaining Rings	NA	0.27	NA	D-19
LATCHES					
10	Drawhook	0.03	0.03	0.06	D-20
11	Spring Clip	0.04	0.03	0.07	D-21
12	Butterfly	0.05	0.05	0.10	D-22
13	ATR (spring loaded, pair)	0.45	0.69	1.14	D-23
14	Lift & Turn	0.03	0.04	0.07	D-24
15	Slide Lock	NA	NA	NA	D-25
TERMINAL CONNECTIONS					
16	Terminal Posts (per lead)	0.22	0.64	0.68	D-26
17	Screw Terminals	0.23	0.45		D-27
18	Termipoint	0.22	0.30		D-28
19	Wirewrap	0.09	0.24	0.14	D-29
20	Taperpin	0.07	0.07		D-30
21	PCB (a) Discretes	0.14	0.17		D-31
22	(b) Flatpacks	0.14 per lead	0.13 per flatpack		D-31
	(c) DIP ICs				
23	• 8 pin	0.46	0.52		D-31
24	• 14 & 16 pin	0.90	0.86		D-31
CONNECTORS					
25	BNC (single pin)	0.07	0.10	10. 17	D-32
26	BNC (multipin)	0.07	0.12	0.19	D-32
27	Quick Release Coax	0.04	0. 04	0.08	D-33
28	Friction Locking	NA	NA	NA	D-34
29	Friction Locking with one Jack Screw	0.18	0.20	0.38	D-35
30	Thread Locking	0.09	0.17	0.26	D-36
31	Slide Locking	0.09	0.12	0.21	D-37

Table D-4. Elemental Maintenance Actions (continued).

Time Standard Number	Description	Standard Times			Reference Fig- ure
		Remove (min.)	Replace (min.)	Interchange (min.)	
32	PLUG IN MODULES DIP ICs (into DIP sockets) CCAs (without tool) (guided)	0.07	0.14	0.21	D-38
33	• 40 pin	NA	NA	NA	D-39
	• 80 pin	0.04	0.07	0.11	D-39
34	CCAs (with tool) (guided)	0.06	0.07	0.13	D-40
35	• 40 pin	0.09	0.08	0.17	D-40
36	• 80 pin	NA	NA	NA	D-41
	CCAs (without tool) (not guided)	0.04	0.16	0.20	D-41
37	Modules	0.09	0.11	0.20	D-42
38	MISCELLANEOUS Strip Wire	-	-	0.10	-
39	Cut Wire of Sleeving	-	-	0.04	-
40	Dress Wire with Sleeving	-	-	0.21	-
41	Crimp Lugs	-	-	0.27	D-43
42	Form Leads (per lead)	-	-	0.03	D-44
43	Trim Leads (per lead)	-	-	0.03	-
44	Adhesives	0.55	0.13	0.68	-
45	Conformal Coating	2.20	0.23	2.43	-
46	Soldering (a) Terminal Post	-	-	0.22	D-45
47	(b) PCB	-	-	0.06	D-46
48	Reflow Soldering	-	-	0.25	-
49	Tinning Flatpacks (dipping)	-	-	0.30	-
50	Desoldering (a) Braided Wick	-	-	0.16	D-47
51	(b) Solder Sucker	-	-	0.09	D-48
52	Form Flatpack Leads (Me- chanically)	-	-	0.11	D-49
53	Clean Surface	-	-	0.29	-
54	Panels, Doors & Covers	0.04	0.03	0.07	D-50
55	Drawers (Large)	0.09	0.10	0.19	D-51
56	Display Lamps	0.10	0.11	0.21	D-52
57	Threaded Connector Cov- ers	0.11	0.14	0.25	-

NOTE: Data obtained from RADC-TR-70-89, Maintainability Prediction and Demonstration Techniques

Table D-5. Common Maintenance Tasks.

Description	Elements of Removal*	Remove (min.)	Elements of Replacement*	Replace (min.)	Interchange (min.)
1. R/R of transistor from a PCB	50(3), 21A(3), 53	1.19	42(3), 21B(3), 47(3), 43(3), 53	1.16	2.35
2. R/R of a transistor from terminal posts	50(3), 16A(3), 53	1.43	42(3), 16B(3), 43(3), 46(3), 53	3.05	4.48
3. R/R of an axial component from a PCB	50(2), 21A(2), 53	0.89	42(2), 21B(2), 47(2), 43(2), 53	0.87	1.76
4. R/R of an axial component from terminal posts	50(2), 16A(2), 53	1.05	42(2), 16B(2), 43(2), 46(2), 53	1.69	2.74
5. R/R of a radial component from a PCB	50(2), 21A(2), 53	0.89	21B(2), 43(2), 47(2), 53	0.81	1.70
6. R/R of a radial component from terminal posts	50(2), 16A(2), 53	1.05	42(2), 16B(2), 43(2), 46(2), 53	1.69	2.74
7. R/R of a terminal point connection	18A	0.22	39, 20B	0.34	0.56
8. R/R of a wirewrap connection	19A	0.09	39, 38, 19B	0.38	0.47
9. R/R of a 16 pin IC from a PCB	50(16), 24A, 53	3.75	24B, 47(16), 43(16), 53	2.59	6.34
10. R/R of a 16 pin flatpack	50(16), 22A(16), 53	5.09	49, 52, 22B, 48, 53	1.08	6.17
11. R/R an 8 pin IC from a PCB	50(8), 23A, 53	2.03	23B, 47(8), 43(8), 53	1.53	3.56

* Numbers in these columns pertain to the time standard numbers in Table D-4. A and B refer to removal and replacement times respectively. The number in parentheses refers to the quantity of each action. R/R = removal and replacement.

Standard Screws

- This time is for all standard threaded fasteners such as; slotted head, Phillips head, and fillister head
- The time given is the time required to remove/replace the fastener from the hole and disengage/or engage it by several twisting motions of the hand (approximately 8 twists)
- Tool required is standard screwdriver (flat head, Phillips, or hex)

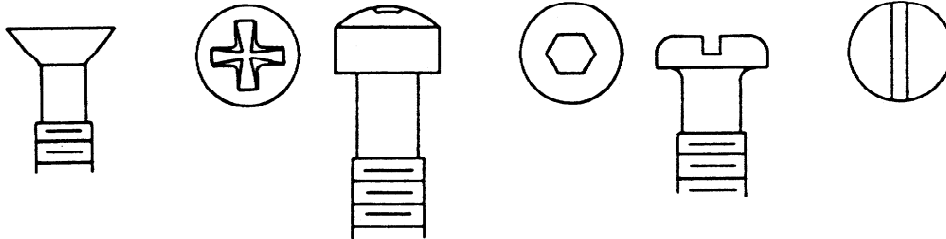


Figure D-11: Standard screws

Hex or Allen Set Screws

- This time is for hex or Allen type set screws
- The time given is for the time to tighten/or loosen a hexagonal type set screw using an Allen type wrench
- Tools required are hex wrenches or Allen type wrenches

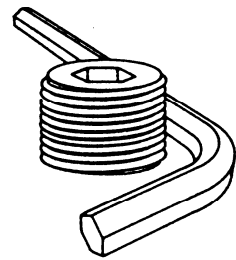


Figure D-12: Hex or Allen set screws

Captive Screws

- This time is for standard fasteners that are captive to the panel/bracket they secure
- The time given for this action includes the time to engage/or disengage the fastener by a series of twisting motions with the hand
- The tool required is a standard screwdriver (flathead, Phillips or hex)

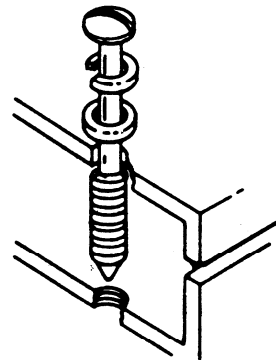


Figure D-13: Captive screws

DZUS Fastener

- This time is for fasteners that require only a 1/4 turn to engage or disengage (fastener is captive)
- The time given is the time required to engage or disengage the fastener by a 1/4 twist motion of the hand
- The tool required is a standard screwdriver (flathead, Phillips or hex)

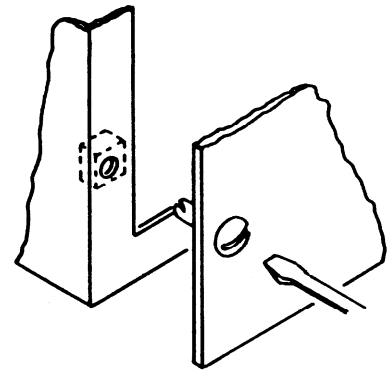


Figure D-14 DZUS fasteners

Tridair Fastener

- This fastener is a quick engaging fastener that requires less than one turn
- This time includes the time necessary to engage/or disengage the fastener using a turn of the hand
- The tool required is an Allen wrench

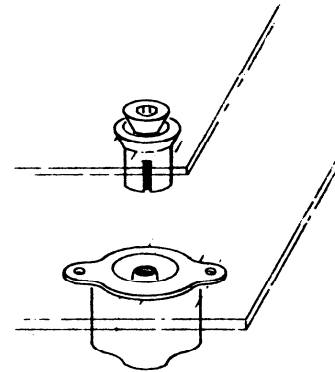


Figure D-15: Tridair fastener

Thumbscrews

- This fastener is a threaded screw with a head that can be grasped easily by the hand
- This time includes the time necessary to disengage/or engage the fastener by turning it with the hand
- No tools required

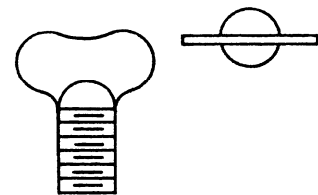


Figure D-16: Thumbscrews

Machine Screws (with nut)

- This fastener is any threaded fastener that does not tap into the structure; instead it engages into a loose nut
- This time includes the time to remove/or position the fastener and nut and the time required to tighten the fastener
- Tools required are a screwdriver and a wrench

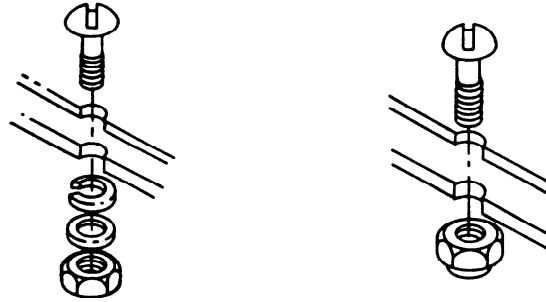


Figure D-17: Machine screws

Nuts or Bolts

- Any fastener that requires a wrench to tighten it down
- This time includes the time necessary to position the wrench and engage/or disengage the fastener

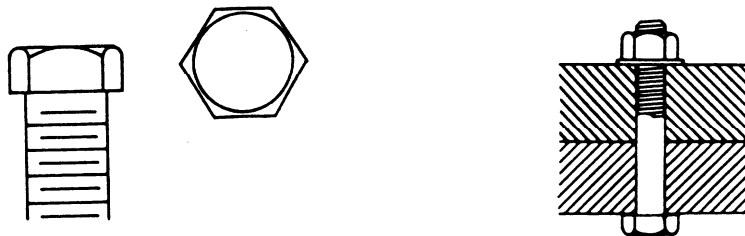


Figure D-18: Nuts or bolts

Retaining Rings

- This device is a "U" shaped piece of metal that retains a unit/component in position
- The time given includes the time necessary to engage/disengage this fastener
- Special pliers are required to remove/replace this fastener

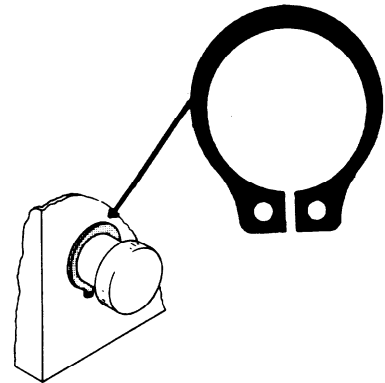


Figure D-19: Retaining rings

Drawhook Latch

- Any latch that is similar to the one shown here
- The time includes the time to engage/disengage the latch completely
- No tools required

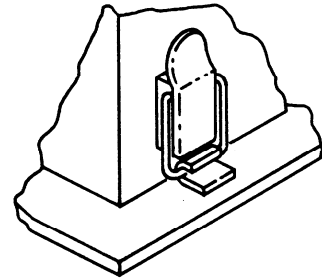


Figure D-20: Drawhook latch

Spring Clip Catch

- Any latch similar to the one shown here
- Time includes the time necessary to engage/or disengage the latch completely
- No tools required

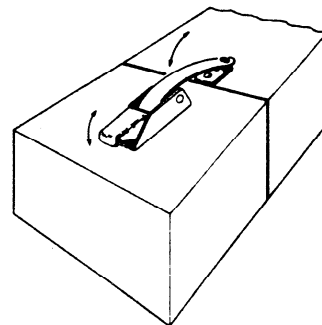


Figure D-21: Spring clip catch

Butterfly Latch

- Any latch similar to the one shown here
- Time includes the time necessary to engage/or disengage the latch completely. Normally consists of the time to lift the tab and turn it 90°
- No tools required

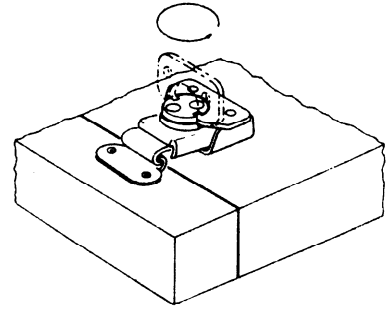


Figure D-22: Butterfly latch

ATR Latch

- Any latch similar to the one shown here
- Time includes the time necessary to unscrew/or screw the cap over the nib to engage/or disengage the secured unit. The time given is for a pair of ATR latches.
- No tools required

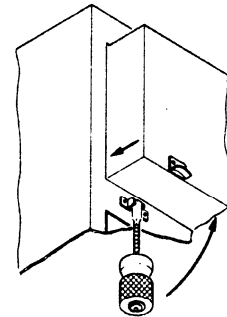


Figure D-23: ATR latch

Lift and Turn Latch

- Any latch similar to the one shown here
- Time includes the time necessary to lift the handle and turn it to unsecure or secure a door or panel
- No tools required

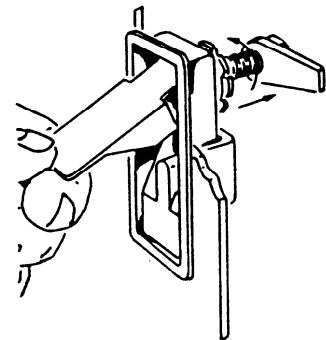


Figure D-24: Lift and turn latch

Slide Lock Latch

- Any latch similar to the one shown here
- The time given is the time necessary to slide the locking device to engage/or disengage the panel
- No tools required

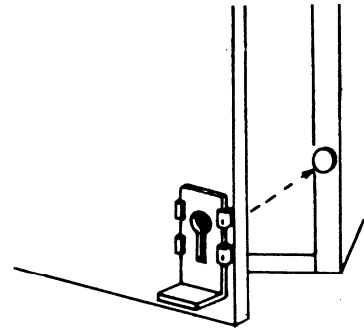


Figure D-25: Slide lock latch

Terminal Posts

- Any terminal connection similar to the ones shown here
- This time is the time required to remove or replace a lead from a terminal (does not include soldering or desoldering)
- Needle nose pliers are required for this task

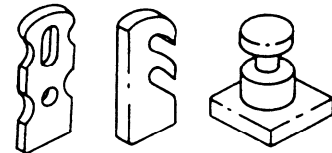


Figure D-26: Terminal post connections

Screw Terminals

- Any terminal connection similar to the one shown here
- This time is the time required to remove/or position the terminal lug and loosen/or tighten the screw
- A screwdriver is required

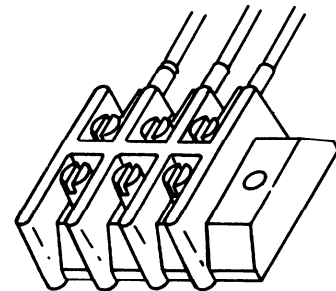


Figure D-27: Screw terminal connections

Termipoint Connections

- Any terminal connection similar to the one shown here
- This time is the time required to remove the clip with a pick or tweezers and the time to replace the clip with a termipoint gun
- Tools required are tweezers, or a pick, and a termipoint gun

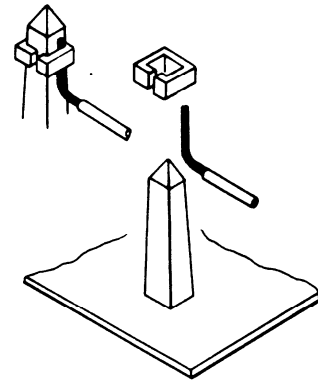


Figure D-28: Termipoint connection

Wirewrap

- Any terminal connection similar to the one shown here
- The time given are to replace the wirewrap with a wire-wrap gun and to remove the connection with an unwrapping tool
- Tools required are a wirewrap gun and an unwrapping tool

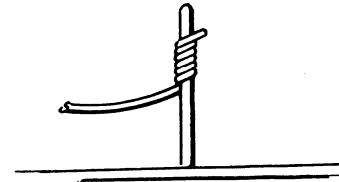


Figure D-29: Wirewrap connection

Taperpin

- Any terminal connection similar to the one shown here
- The time given is the time required to unplug mate or demate the connector
- No tool required

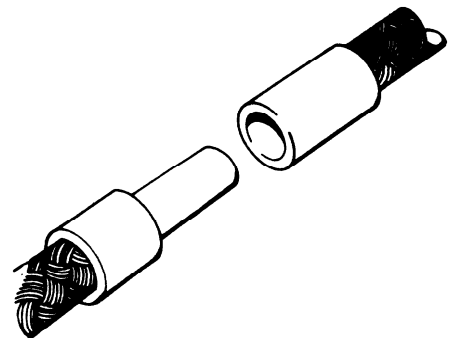


Figure D-30: Taperpin connection

PCB

- Any terminal connected directly to the printed circuitry of a circuit card
- The time given is the time required to remove or replace a lead from the PCB (no soldering or desoldering time included)
- The tools required are a pick or needlenose pliers

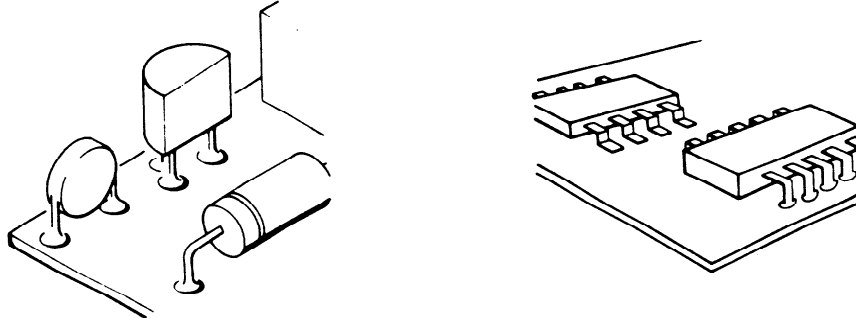


Figure D-31: PCB connections

BNC Connectors

- Any connector that has a bayonet-locking device
- Times given are for mating/demating the connectors by a twisting motion
- No tool required



Figure D-32: BNC connectors

Quick Release Coax Connectors

- Any coax connector that engages or disengages by a push or pull motion
- Times given are for demating/mating the connectors by a pulling or pushing motion
- No tool required

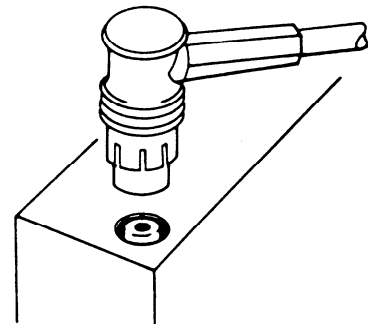


Figure D-33: Quick release coax connectors

Friction Locking Connectors

- Any connector that is mated and secured by the friction of the pins and/or connector case
- The time given is the time necessary to mate or demate these connector types
- No tool required

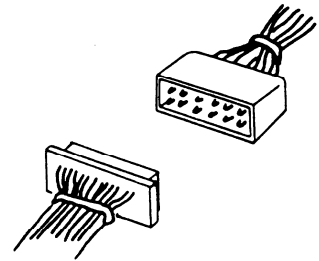


Figure D-34: Friction locking connector

Friction Locking Connector with Jackscrew

- Any friction locking connector that is secured by a jack-screw
- The time given is the time necessary to demate/mate the connector by disengaging/engaging the jackscrew
- No tool required

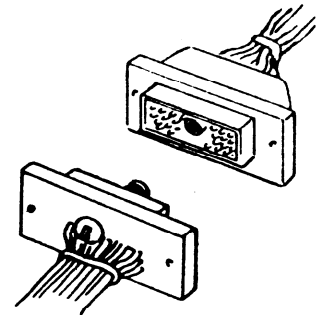


Figure D-35 Friction locking connector with jackscrew

Thread Locking Connector

- Any connector that is secured by a threaded connector shell
- The time given is for the demating/mating of the connector and the securing/unsecuring of it by a turning motion
- No tools required

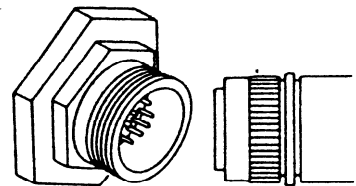


Figure D-36: Threadlocking connector

Slide Locking Connector

- Any connector that is secured by a slide lock
- The time given is the time required to mate/demate the connector and engage/disengage the slide lock
- No tools required

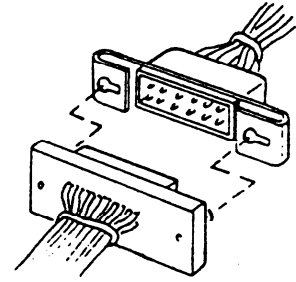


Figure D-37: Slide locking connector

DIP ICs

- This includes any dip IC that is secured in a dip socket
- The time given is the time required to unplug or plug in the dip IC
- No tools required

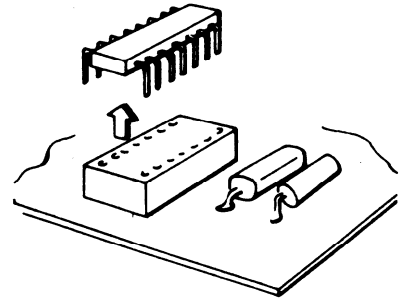


Figure D-38: Dip ICs

Guided CCAs

- Any guided CCA that is inserted/removed by hand
- The time given is the time to pull out or push in the CCA
- No tools required

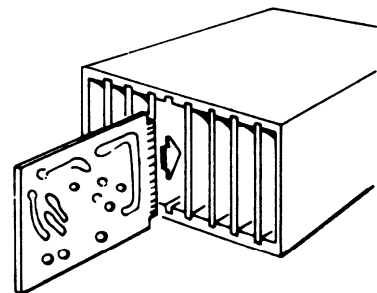


Figure D-39: Guided CCAs

Guided CCAs with a Tool

- This time is associated with any CCA that is inserted/removed with a card extracting tool
- The time includes the time required to position the tool and remove/replace the CCA
- A card extracting tool is required

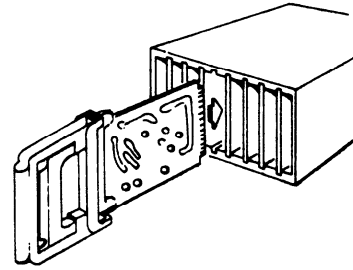


Figure D-40: Guided CCAs with a tool

Non-guided CCAs

- This time is associated with plug-in cards that are not guided
- The time includes the time required to remove/replace the CCA from the edge connector (does not include time for fasteners)
- No tools required

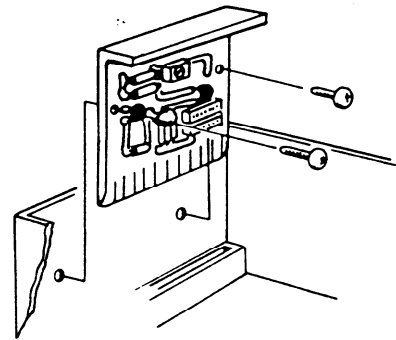


Figure D-41: Non-guided CCAs

Modules

- This is the time associated with removing or positioning a modular assembly
- The time is the time necessary to remove the module or position it in place
- No tools required

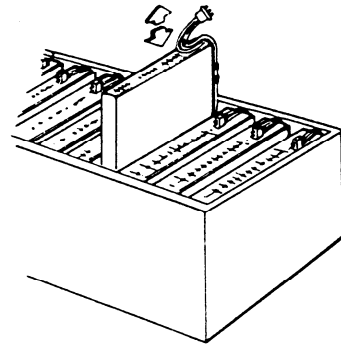


Figure D-42: Modules

Crimp Lugs

- This is the time associated with securing a terminal lug to a wire
- The time given includes the time to position the wire in the lug and crimp it
- A crimping tool is required

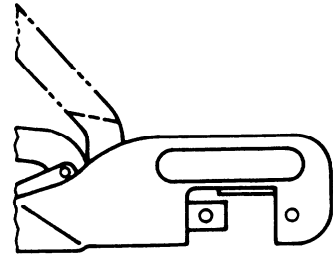


Figure D-43: Crimp lugs

Form Leads

- This is the time associated with forming a lead on a component prior to connecting it to a terminal
- The time given is the time necessary to grasp the lead with the pliers and form it
- Needle nose pliers are required

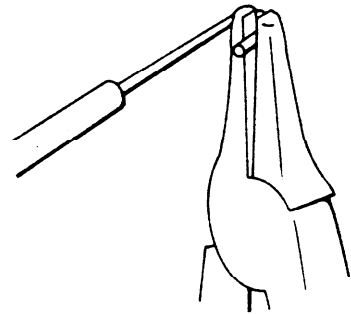


Figure D-44: Form leads

Soldering Terminal Posts

- The time associated with soldering a lead to a terminal post
- The time given is the time to heat the terminal post and apply the solder
- A soldering iron is required

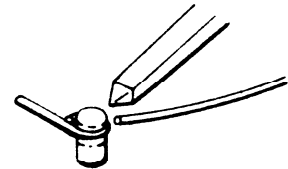


Figure D-45: Soldering terminal posts

Soldering PCB Connections

- The time associated with soldering a lead to a PCB
- etching
- The time given is the time to heat the etching pad and apply the solder
- A soldering iron is required

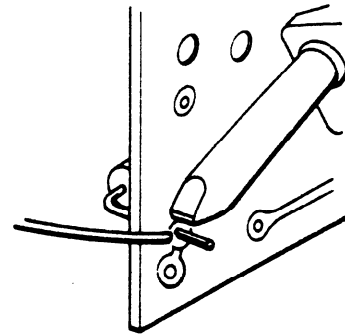


Figure D-46: Soldering PCB connections

Desoldering using a Braided Wick

- The time associated with desoldering a connection using a braided copper wick
- The time given is the time to reheat the solder and extract it from the PCB or terminal using a braided copper wick
- A soldering iron and braided copper wick are required

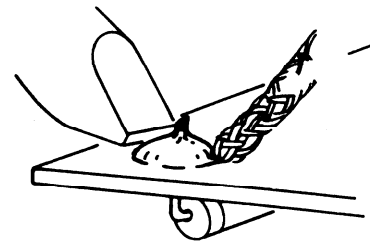


Figure D-47: Desoldering with a braided wick

Desoldering using a Vacuum

- The time associated with desoldering a connection using a vacuum assisted desoldering iron
- The time given is the time required to reheat and "suck-up" the solder
- A desoldering iron is required

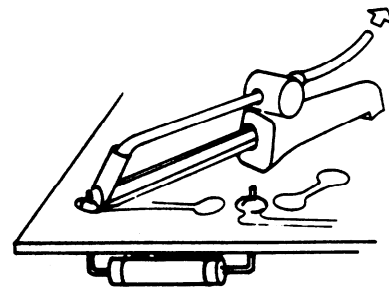


Figure D-48: Desoldering using a vacuum

Form Flat Pack Leads

- The time associated with forming flatpack leads using a mechanically operated die
- The time given is the time required to position the flat-pack and actuate the mechanism
- A mechanically operated device is used to do this

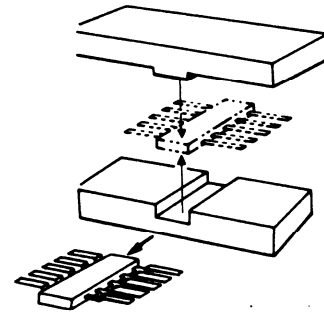


Figure D-49: Form flat pack leads

Panels, Doors, and Covers

- The time associated with open/closing panels, doors and covers
- No tools required

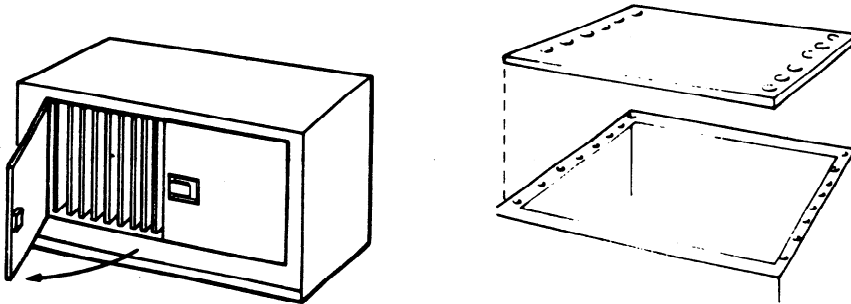


Figure D-50: Panels, doors and covers

Drawers

- The time associated with opening/closing of drawers that are on a track
- No tools required

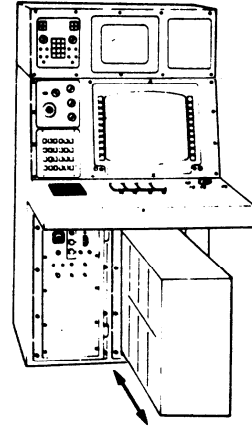


Figure D-51: Drawers

Display Lamps

- The time required to remove/replace panel indicators that pop in and out
- No tools required

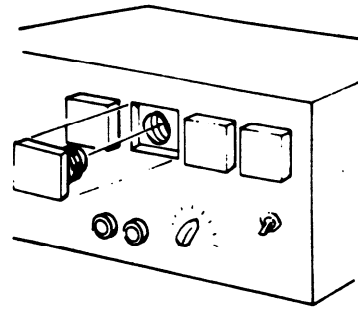


Figure D-52: Display lamps

ANNEX B: RESULT OF DB AG STUDY - REQUIREMENTS FOR AVAILABILITY OF ERTMS EQUIPMENT - CONTENTS .

I. References

- | | |
|---|--|
| 1. prEN50126 | Railway applications - The specification and demonstration of Dependability, Reliability, Availability, Maintainability and Safety (RAMS)“ CENELEC TC9X, Draft November 1995 |
| 2. IEC 605-1 | Equipment reliability testing; part 1: General requirements |
| 3. IEC 605-7 | Equipment reliability testing; part 7: Compliance test plans for failure rate and mean time between failures assuming constant failure rate |
| 4. IEC 1025 | Fault tree analysis (FTA) 1st edition |
| 5. Dr. Wegel, Dr. Rhein, Newi, Wiemann, Mihm, ... | Technical and operational availability of the ERTMS-System |

1. General remarks

In general failures in the railway control- and signalling systems will effect the operation e.g. visible for the railway user in form of train delays or invisible e.g. higher costs for providence of rolling stock or infrastructure.

To reduce the effects of failure and to avoid overengineering of technical equipment it is necessary for the railway company to know the different effects in case of failure of railway components, especially for a railway operator it is more and more essential to know the amount of train delay which will derive from a failure of a new system, e.g. of a new signalling system.

In the following there will be presented a new method based on a study carried by DB-AG and partners to define the necessary technical availability of the ERTMS-System in accordance to the operational needs of a railway company.

2. Description of operational conditions

2.1 Definition of delay classes

Technical or operational failures of railway equipment can effect in different manner the train operation. Depending of this effects, especially of the degree of train delay the railway operator will have to adopt measures to ensure a safe and as quite as possible comfortable service. For registration, static and also to specify the necessary measures for a good service train delays can be classified in five categories.

Class of delay	Delay time
1.	1 to 5 min.
2.	6 to 15 min.
3.	16 to 30 min.
4.	31 to 60 min.
5.	> 61 min. ³

Assuming a linear distribution the average delay of a train will be expected in the middle of the above defined delay class:

Class of delay	Delay time	Average delay
1.	1 to 5 min.	3 min.
2.	6 to 15 min.	10,5 min.
3.	16 to 30 min.	23 min.
4.	31 to 60 min.	45,5 min.
5.	> 61 min. ⁴	75,5 min.

³ for better calculation of average train delays this class will be defined as 61 to 90 min.

⁴ for better calculation of average train delays this class will be defined as 61 to 90 min.

Considering the occurrence of train delay within the different classes the resulting average train delay will be:

$$t_{vm} = \sum_{v=1}^5 a_v b_v = (3 * a_1 + 10,5 a_2 + 23 * a_3 + 45,5 A_4 + 75,5 a_5)$$

with t_{vm} = average train delay
 a_v = probability of train delay in class v

2.2 Operational requirements

To assure a sufficient quality of operating services, the admissible delay of trains due to operational and/or technical failures will be defined as 5% of all trains having a delay ≥ 5 min.. From this margin only 10 % will be admissible for the control- and signalling system. From the remaining admissible limit of 0,5% about 40 % will be reserved for the ERTMS-system:

Delay class	Admissible limit of train delay due to failures of the ERTMS-system
1 - 5 min.	1,24 %
6 - 15 min.	0,14 %
16 - 30 min.	0,04 %
31 - 60 min.	0,02 %
> 60 min.	0,001 %

Additionally to the definition of limits of delay the total number of delay minutes within a defined railway organisation has to be reduced.

3. Consideration of used components for the ERTMS-System

To define the average delay time for long distance trains there will be considered the following configuration for an average trip of 500 km:

Table 1: Network configuration				
	Distance between components	Number of elements met during one trip $n_{E/Z}$	mean down time MDT_E	Influence factor k_E
Locomotives		1,25 ⁵	4	2,5
Radio equipment Base transmission stations (BTS)	every 7 km	71,43	4	72,68
Radio Block Centre (RBC)	every 50 km	10	2	12,5
Balise	every 1 km	500	24	500,2
ERTMS-Central	every 50 km	10	2	12,5

The influences to operation due to a failure of one of this five components depends not only of the number of elements met during a trip but also of the mean down time (MDT) for this components. The effects of different MDT of the above mentioned elements will be considered by the definition of the influence factor k_E :

$$k_E = (n_{E/Z} * MDT_E + T_F) / MDT_E$$

with MDT_E = mean down time of element E
 T_F = average trip duration

The factor k_E indicates in which degree the nonavailability of a component E will affect the train operation. Considering this factor the probability of operation disturbances Q_Z for one trip can be defined:

⁵ due to changes of 25% of locomotives for operational needs in terminals

$$Q_Z = 2,5 U_{OBE} + 72,68 * U_{BTS} + 12,5 * U_{RBC} + 500,2 * U_{Bal} + 12,5 * U_{CE}$$

with U_{OBE} = Unavailability of Onboard equipment
 U_{BTS} = Unavailability of Base transmission station (Radio)
 U_{RBC} = Unavailability of Radio Block Center
 U_{Bal} = Unavailability of Balise
 U_{Ce} = Unavailability of ERTMS Central

The ERTMS system itself will be composed by the following components:

Table 2: Components used for the ERTMS System	
I) Onboard equipment:	
1.	EVC (European vital computer)
2.	Odometer
3.	Radio mobile station
4.	Baliseinterrogator
5.	Man Machine Interface (MMI)
6.	Train Interface Unit (TIU)
II) Infrastructure (fixed elements):	
7.	Balise
8	RTC-Centre/RTC function including S_{2m} interface - (connection to Radio equipment, terminal, central computer,....)
9	RTC-Centre /Interface route protection
III) additionally for radio transmission:	
10.	Radio equipment - Base transmission stations
11.	Radio block centre (RBC)

4. Results of the FMEA Analysis

According to FMEA technics in the following tables 3 and 4 have been considered the effects of failures due to different components for performing the ERTMS system.

In column 3 of this tables are indicated the main functions of this components. Columns 4 and 5 of this tables are reserved in case for further studies (realised by national railways together with industry) to discuss different reasons for the disturbance of functions / components. Also column 6 serves for a further discussion to describe different effects of failure in separate phases of a trip. For the definition of the necessary overall technical availability of this components there will only be considered the effect of a failure, but not distinguished between different operational situations or reasons of disturbances. This can be done in future on basis of practical experiences e.g. after realisation of different installations of the ERTMS system to improve the definition of requirements of system components. In column 7, 8 and 9 are described the operational effects of the failures e.g. column 7 indicates if it is necessary to brake, 8 the conditions of trip continuation and 9 the necessary operation measures. In column 10 to 13 have been indicated the possible trip delay due to one failure of one of this components, e.g. for the balises the figure 0,1 indicates that (a maximum of about) 10 % of all the trains will have a delay of about 1-5 minutes due to a failure of one balise. Column 15 describes the assumptions made e.g. to define the probable train delay and column 16 indicates advices for the planning of a ERTMS system.

In table 5 there has been effected the calculations of probable delay time for the defined delay classes. in column 18 (but also in columns 7, 9, 11, 13 and 15 there can be seen the influence of a component to the whole delay time (e.g. balise interrogator - 30,59% of the whole delay time). These column can be used to decide what component shall be made redundant to reduce efficiently the delay time. In columns 6, 8, 10 12, 14 and 16 can be seen the value of a possible train delay due to a failure of one component. Column 17 indicates the summarised delay contribution in hours of one component. In summary at the end of column 17 there is given the overall delay time for the considered ERTMS system.

Analysis of the operational effects of potential interruptions in the radio train control (RTC) system (FMEA)

table 3

No.	Failed component	Failed function	Cause of interruption	Interruption identification	Operational situation	Operational effects						Assumptions and pre-conditions	Conclusions and advice		
						Brake to	Continue with	Action	Delay minutes Frequency ratio						
									0-5	5-15	15-30	>30	Mean		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	ST, balise	Balise telegrams	all	yes	any	-	RTC	None	0.1				0	Detection tolerates failure of one balise	Observe balise configurations before danger points
2	Two adjacent balises	Balise telegrams	all	yes		0 to dang. pt.	RTC or V100	Braking to danger point, continuation after reception of first balise possible, otherwise continuation on command	0.7	0.3			5	Fault indication, failure of balises limited locally	Check simplified operations procedures
3	Vehicle equipment, reliable computer	Safety	all	yes		0	V100	Automatic brake application to HALT, disconnect RTC, continuation on command			0.2	0.8	41	Fault indication	Change of locomotive if necessary
4	Vehicle equipment, odometer	Internal distance measurement	all	yes		0	dep. on driver ($\leq V100$)	Brake application to HALT, continuation on command, or change of locomotive			0.2	0.8	41	Fault indication	Independent speed indicator important
5	Vehicle equipment, mobile station	Radio interface	all	yes		0 to dang. pt.	V160 ... V100 V40	Completion of running permission available, continuation on command or run-ning at sight			0.3	0.7	38	Fault indication, running permission available may be completed	Redundant train radio is important, change of locomotive if necessary
6	Vehicle equipment, balise interrogator	Reception, balise telegrams	all	yes		0 to dang. pt.	V100 (160?)	continuation on RTC command till admissible tolerance of internal distance measurement (stop), maximum till next danger point, continuation on command			0.5	0.5	34	Fault indication, running permission available may be completed	Change of locomotive if necessary
7	Vehicle equipment, MMI	Cab signalling, inputs driver	all	yes		0	V100	Braking to V160, completion of running permission available, continuation on command or change to automatic			0.2	0.8	41	Fault indication, redundant train radio available	Independent speed indicator important

Analysis of the operational effects of potential interruptions in the radio train control (RTC) system (FMEA)

Table 4

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
8	Vehicle equip-ment, TIU	Brake control	all	yes (safe reaction)		0	V100	Automatic brake application to HALT, continuation on command, change of locomotive			0.2	0.8	41	Fault indication, close air shut-off cock	
9	Vehicle equip-ment, power supply	Power supply	all	yes		0	V100	Automatic brake application to HALT, continuation on command, change of locomotive			0.2	0.8	41	Fault indication	Speed indicator independent of power supply necessary, redundant train radio
10	BTS	Radio supply vehicle	all	yes	Next danger point in the cell or in the adjacent cell	0 to dang. pt.	V100 or RTC	Braking to danger point (completion of running permission), running on command in the area of the adjacent BTS and/or continuation with RTC after obtaining radio contact with adjacent BTS	0.5	0.2	0.3		10	Running permission available may be completed	Redundant illumination in area of danger points, redundant train radio, subsidiary signals (permit V100)
11	BSC	Transmission system	all	yes		0 to dang. pt.	V100	completion of running permission till danger point, continuation on command	0.2	0.6	0.2		11	Fault indication, running permission available may be completed	Redundant train radio, subsidiary signals necessary
12	MSC	Transmission system	all	yes		0 to dang. pt.	V100	completion of running permission till danger point, continuation on command				1.0	75	Running permission available may be completed	Redundant train radio, subsidiary signals necessary
13	Transmis-sion/ region	Transmission system	all	yes		0 to dang. pt.	V100	completion of running permission till danger point, continuation on command		0.3	0.7		19	Running permission available may be completed	Redundant train radio, subsidiary signals necessary
14	Transmis-sion/ region	Transmission system	all	yes		0 to dang. pt.	V100	completion of running permission till danger point, continuation on command				1.0	75	Running permission available may be completed	Redundant train radio, subsidiary signals necessary
15	RTC centre	RTC function including S _{2m} interface	all	yes		0 to dang. pt.	V100	completion of running permission till danger point, continuation on command		0.3	0.7		19	Running permission available may be completed	Redundant train radio, subsidiary signals necessary
16	RTC centre	Interface route protection	all	yes		0 to dang. pt.	V100	braking to danger point, continuation on command		0.3	0.7		19	Running permission available may be completed (signal box must not alter route)	Replacement operation RTC centre by redundant communication if necessary

Delay contribution of ERTMS-components in case of failure - Delay distribution to the defined Delay classes

Table 5

No	Component	Unavailability	Factor	Delayed train to delay class	Delay contribution
----	-----------	----------------	--------	------------------------------	--------------------

		U _e	k _e	k _e *U _e	1-5 Min.	%	5-15 Min.	%	15-30 Min.	%	31-60 Min.	%	61-90 Min.	%	Total 6-14	Minutes	%
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	ST, Balise	1,00E-06	500,2	5,00E-04	5,0 E-05	6,37%									5,00E-05	0,00015	0,27%
2	Two adjoined balises	1,00E-12	500,2	5,00E-10	3,5 E-10	0,00%	1,5 E-10	0,00%							5,00E-10	0,00015	0,00%
3	OBE, EVC	2,00E-06	2,5	5,00E-06					1,00E-06	0,10%	4,00E-05	0,74%			5,00E-06	0,00021	0,37%
4	OBE, Odometer	2,00E-05	2,5	5,00E-05					1,00E-05	0,97%	4,00E-05	7,37%			5,00E-05	0,00205	3,66%
5	OBE, mobile interface	2,50E-06	2,5	6,25E-06					1,88E-06	0,18%	4,38E-06	0,81%			6,25E-06	0,00024	0,43%
6	OBE, Baliseinterrogator	2,00E-04	2,5	5,00E-04					2,50E-04	24,31%	2,50E-04	46,09%			5,00E-04	0,01713	30,59%
7	OBE, MMI	2,50E-08	2,5	6,25E-08					1,25E-08	0,00%	5,00E-08	0,00%			6,25E-08	0,00000	0,00%
8	OBE, TIU	1,20E-04	2,5	3,00E-04					6,00E-05	5,83%	2,40E-04	44,25%			3,00E-04	0,01230	21,97%
9	OBE, power supply ⁶	2,00E-06	2,5	5,00E-06					1,00E-06	0,10%	4,00E-06	0,74%			5,00E-06	0,00021	0,37%
10	BTS	2,00E-05	72,7	1,45E-03	7,3 E-04	92,65%	2,9 E-04	68,33%	4,36E-04	42,41%					1,45E-03	0,01527	27,27%
11	BSC	4,00E-06	9,6	3,84E-05	7,7 E-06	0,98%	2,3 E-05	5,41%	7,68E-06	0,75%					3,84E-05	0,00044	0,79%
12	MSC	1,00E-06	4	4,00E-05									4,00E-06	36,36%	4,00E-06	0,00030	
13	Cable BTS-BSC	2,00E-05	5,5	1,10E-04			3,3 E-05	7,75%	7,70E-06	7,49%					1,10E-04	0,00211	
14	Cable BSC-MSC	2,00E-06	3,5	7,00E-06									7,00E-06	63,64%	7,00E-06	0,00053	
15	RTC-Centre RTC function including S _{2m} interface	1,00E-06	12,5	1,25E-05			3,8 E-06	0,88%	8,75E-06	0,85%					1,25E-05	0,00024	0,43%
16	RTC-Centre Interface route protection	2,00E-05	12,5	2,50E-04			7,5 E-05	17,62%	1,75E-04	17,01%					2,50E-04	0,00481	8,60%
	Total 1-16				7,9 E-04	100%	4,3 E-04	100%	1,03E-03	100%	5,42E-04	100%	1,1E-05	100 %	2,79E-03	0,05599	100,00%
													probability of train delays: average delay: probability of train delay > 5 Min.:				0,28% 20,1 Min. 0,20%

⁶ power supply of components for the ERTMS system

5. Definition of the necessary technical availability

Considering the state of the art of technics and also availability figures for similar technics there will be defined (as maximum!) the following values for the unavailability of components used for the ERTMS system to achieve the defined operational requirements. The unavailability of this components has to be calculated according to the respective IEC and MIL standards. In case that the indicated unavailability of a component will not be reached the supplier has to show - using the presented method - that the overall availability of the system will be ensured.

Table 2: Unavailability of components used for the ERTMS system:		
Subsystem	Component	Unavailability
Onboard equipment	OBE, EVC	$\leq 2,0 \cdot 10^{-6}$
	OBE, Odometer	$\leq 2,0 \cdot 10^{-6}$
	OBE, Mobilstation	$\leq 2,5 \cdot 10^{-6}$
	OBE, Baliseinterrogator	$\leq 2,0 \cdot 10^{-4}$
	OBE, MMI	$\leq 2,5 \cdot 10^{-8}$
	OBE, TIU	$\leq 1,2 \cdot 10^{-4}$
Infrastructure	Balise	$\leq 1 \cdot 10^{-6}$
	RTC-Centre	$\leq 1 \cdot 10^{-6}$
	RTC function including S _{2m} interface	$\leq 2 \cdot 10^{-5}$
	RTC-Centre Interface route protection	$\leq 2 \cdot 10^{-5}$
Radio equipment	RBC	$\leq 4 \cdot 10^{-6}$
	BTS	$\leq 2 \cdot 10^{-5}$

6. Further aspects and results of the FMEA analysis

Besides the definition of technical availability due to a prescribed aim of operational availability (prescribed admissible delay target) the described method for the FMEA-Analysis can also be used to discuss the different effects of redundancy of components, e.g. reduction of delay time due a failure of the MMI by using a redundant display, or to define the influence of a degraded mode/fall back level.

There is also possible to calculate the overall delay time in minutes for a defined railway system like e.g. long distance trains, suburbs trains, highspeed lines with a certain number of trains per day.

In case that the operational requirements will not be met by the used components tables 3 and 4 can be used also to check for the possible reasons of non-compliance e.g. loss time for operational measures greater than stated in table 3 or 4.